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A HISTORICAL OVERVIEW OF THE PHILOSOPHY BEHIND KEYBOARD FINGERING INSTRUCTION FROM THE SIXTEENTH CENTURY TO THE PRESENT

By

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ABSTRACT

Fingering has been among the most controversial aspects of keyboard instruction, as it is one of the hardest to approach and systematize due to the individuality of the human hand and the multiple fingering combinations that could apply to each musical passage. Keyboard instruction books, since their first appearance in the sixteenth century, have made a continuous and systematic attempt to provide students and teachers with advice and methodologies on fingering. A historical overview of keyboard fingering instruction material of the last five centuries reveals a remarkable diversity, due not only to the various degrees of systematization of the instructional material, but also to the fundamentally different pedagogical methodologies. These differences are associated with the keyboard repertoire, performance practices, and historical and sociological contexts of different eras. Pre-Baroque and Baroque paired keyboard fingering was based on articulation demands of the keyboard repertoire of the time, which was meant to be performed predominantly on organ and harpsichord. The period of Enlightenment initiated a fundamental change in both the level of organization of the keyboard teaching material and the specific fingering instructions. Treatises of that time reflect a transition from the Baroque to the Classical style, instruments and technique. The advantage of giving a pivoting role to the thumb, that became a standard performance practice, changed the philosophy of teaching fingering. Teachers from the first half of the nineteenth century aspired to provide piano students with a technical system that would ensure the utmost finger dexterity. The output of "finger gymnastic" material has trained generations of pianists; however, soon after its conception this material was heavily criticized for its mechanical and anti-artistic character. As a result, the Romantic philosophy of teaching fingering revolved around tone production and special timbral effects. Scientific achievements in the fields of anatomy and physiology were the basis for the bulk of twentieth-century keyboard instruction books. Instead of prescribing fingering formulas and rules, the contemporary pedagogical norm is an awareness of the complexity of the playing apparatus that enables the pianist to make fingering decisions

based on individual muscular abilities, tension and relaxation issues, and desired tone colors.

INTRODUCTION

Since their first appearance in the sixteenth century, keyboard instruction books have made a continuous and systematic attempt to provide students and teachers with an organized presentation of various issues related to keyboard playing. The variety of these issues is overwhelming, not only because of the idiosyncratic nature of teaching a musical instrument, but also because of the different musical performance practices in different eras and geographical places.

As a result, manuals of this sort include theoretical information, advice, and examples of the skills that a keyboardist is expected to master depending on the musical standards of each period. For instance, tutors from the Baroque era include lengthy analysis of thoroughbass principles, while more recent books elaborate on appropriate stylistic approaches to pieces from various periods.

Despite the enormous differences that can be observed in keyboard instruction books in terms of origin, style, language, and organization, they all have one common source: the desire of experienced teachers to summarize years of knowledge. Thus they provide young musicians with essential reference tools to help them master the technical and interpretive aspects of keyboard playing.

Of all these aspects, fingering has been the most controversial, since it is perhaps the hardest to approach and systematize due to the individuality of the human hand and the multiple fingering combinations that could apply to each musical passage. In addition, for every rule, an infinite number of exceptions could be pointed out based on the musical context that precedes and follows each given example.

Despite the controversial nature of the topic, fingering instructions are included in almost every keyboard manual ever written. Furthermore, published exercises, etudes, or even performance pieces that include fingering suggestions made by composers or famous teachers provide additional information on practices from different eras. All of this material reflects an enormous diversity of approaches to fingering based on the progress of scientific anatomical knowledge, the evolution of keyboard instruments, the

individual technical demands of repertoire from different periods, advances in piano pedagogy, and various performance practices as they were applied in different regions and eras.

Information derived from fingering sources in regard to articulation, phrasing and music interpretation has been a source of interest for many musicological and performance practice analyses. In particular, treatises whose content and organization changed the course of systematic piano pedagogy perception, such as Carl Philipp Emanuel Bach's *Versuch über die wahre Art das Clavier zu spielen,* have undergone extensive research by succeeding generations.

Nevertheless, an overview of the changes in fingering instruction throughout the centuries has not been undertaken. In addition, the vast majority of existing research focuses on subjects related to performance practice issues, more specifically to the direct relationship between fingering and the idiosyncratic character of the music to be performed. This historical overview of fingering resources will highlight the major scientific, sociological, pedagogical and musical reasons behind the philosophical and practical instructional differences. It will also examine the principles of fingering that have prevailed throughout the centuries, whether referring to specific rules or to general goals.

The present study focuses on four separate and distinct periods. The first period will include keyboard fingering material written from approximately 1520 to 1750, with particular emphasis on treatises from different geographical regions. The second period will cover the transition from harpsichord and organ playing to the predominance of fortepiano, covering the years between 1750 and 1840. The third period's development of the modern piano and increased requirement for virtuosity generated a need for unprecedented finger dexterity, coinciding roughly with the Romantic era; the discussion will cover treatises and teachings from 1840 until 1900. Finally, the scientific approach to piano pedagogy that derived from the knowledge of motor skills, as well as the use of "unconventional" piano techniques and its application to fingering, will be the basis for the final period, beginning with the turn of the twentieth century and extending to the present time.

CHAPTER 1

RENAISSANCE AND BAROQUE PERIODS

SOURCES FROM 1520 TO 1650

Introduction

The humanistic spirit that prevailed throughout the Renaissance was the driving force behind all the major scientific and artistic developments from the fourteenth century through the sixteenth. The pursuit of a higher reality and the replacement of authority by empiricism produced an era of intense scientific observation and an artistic desire to create order.[1] Johannes Gutenberg's invention of the printing press, Leonardo da Vinci's numerous manuscripts—such as the Codex Leicester, a revolutionary writing on astronomy—and Columbus's discovery of America are only a few of the scientific achievements of the time.

The arts were certainly not unaffected by the quest for advancing the human intellect. "Art" music in particular experienced the beginnings of disassociation from its strictly religious character. Demand for secular music increased, while advancements in instrument making accelerated. By the late sixteenth century composers were able to write idiomatically for instruments with a gradual abandonment of vocal compositions as instrumental prototypes.[2] Keyboard compositions included canzonas, ricercars, toccatas, dance variations, and other short forms.

As the keyboard repertoire expanded and the mechanics of the instrument constantly improved, the demand for keyboard instruction began to emerge. This need, in accordance with the Renaissance ideal of a solid educational system, resulted in the production of numerous treatises on music. Even though theoretical music writings had

[1] Douglass Seaton, *Ideas and Styles in the Western Musical Tradition* (Mountain View: Mayfield Publishing Company, 1991), 94.

[2] Donald Jay Grout, *A History of Western Music* (New York: W.W. Norton & Company, 1973), 276.

been in existence since antiquity, it was not until the sixteenth century that instruction tutors appeared for the first time.

A prominent characteristic of keyboard instruction material from its beginning and throughout the Baroque period is the remarkable diversity observed in fingering instructions. Even though the philosophy of fingering throughout Europe was based on the unequal length and strength of fingers, treatises provided multiple answers to the question of which fingers are actually stronger, even though avoidance of the thumb and the little finger seems to be widely accepted. The considerable differences between fingering systems underline the individuality of performance practices and the existence of distinctive national styles.

The roots of advanced nationalism in Europe in the sixteenth century could be attributed at least partially to the Reformation and the political oppression that caused the fragmentation of the Roman Catholic Church. For the history of music this "meant the growth of a variety of practices and musical styles and repertoires."[3]

The bulk of keyboard tutors from 1520 to 1650 came from Germany, Spain and Italy. Despite the extraordinary flowering of the variation form in England in the late sixteenth and early seventeenth centuries, the English sources of the time do not include any pedagogical discussion. However, there is an abundance of fingerings indicated in the so-called "virginal" music found in publications and manuscripts.

France experienced perhaps the most isolated and independent musical development throughout the Baroque period. During the sixteenth century the religious wars between Catholics and the Calvinist Huguenots prevented a significant artistic development; the very first French harpsichord tutor appeared as late as the beginning of the eighteenth century. Even collections of pieces with performance indications did not appear in France until 1665.[4]

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[3] Douglass Seaton, 134.

[4] Cynthia Qualls Ashley, "An Examination of Early Keyboard Fingering with Emphasis on the Development of National Styles" (Creative Project Paper, Southeast Missouri State University, 1987), 69.

German Sources

Hans Buchner von Constanz's (1483-1540) *Abschrifft M. Hansen von Constanz, des wyt [sic] Beriempten Organisten Fundament Buch sinen [sic] Kinden Verlosse* is the earliest source of information on keyboard fingering. The tutor, written probably around 1526, was actually prepared by Christoph Piperinus in 1551 and has survived in three manuscripts in both Latin and German.

Buchner's tutor is comprised of three chapters and a comprehensive set of liturgical compositions. In the first chapter, the author includes a set of rules for fingering and a thoroughly fingered three-voice hymn setting in German organ tabulature as an example of fingering applications. In the introduction Buchner acknowledges the complexity of providing specific fingering instructions because of the number of possible exceptions. Nevertheless, he considers the matter of utmost importance:

> Unless every note is taken with its appropriate finger, many [virtues] are lost in playing, which if they are present, bring to the melody a wonderful grace and joyfulness.[5]

In the examples given in *Fundamentum*, the common reference name for the tutor, use of the thumb and the little finger is avoided. Buchner, like many authors of early keyboard methods, uses a system of numbering the fingers which is different from the "modern fingering system."[6] In *Fundmentum* the thumb is numbered as "5" and the remaining fingers from the index through the little one are numbered as "2" to "4" respectively. With few exceptions Buchner follows the principle of using the second and fourth fingers on the beat notes and the third finger on the off-beat notes.[7]

While the given rules do not include detailed information about the crossing of one finger over the other, an analysis of the given examples results in hand positions which are rather unconventional by today's standards. Buchner is careful to specify that

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[5] Hans Buchner von Constanz, *Abschrifft M. Hansen von Constanz, des wyt [sic] Beriempten Organisten Fundament Buch sinen [sic] Kinden Verlosse,* 1551; trans. Mark Lindley in *Ars Ludendi: Early German Fingerings c.1525-c.1625* (Neuhof: Tre Fontane, 1993), 42.

[6] For the remaining of the current thesis the term "modern fingering system" will refer to the contemporary commonly used numbering system for fingers. This system assigns the numbers "1" to "5" to fingers, beginning with the thumb as "1" and ending with the little finger as "5" in both hands.

[7] Newman Wilson Powell, "Early Keyboard Fingering and its Effect on Articulation" (Master's Thesis, Stanford University, 1954), 12.

when performing the interval of a third with the right hand, the fourth finger (in modern fingering system) takes the upper note and the second the lower.

Newman Wilson Powell in his thesis on "Early Keyboard Fingering and its Effect on Articulation" observes:

> It would hardly have seemed necessary for him to clarify this point unless the arm was frequently held in a position that would make the opposite disposition of fingers at least possible.[8]

Buchner's tutor may lack detail on the subject; nevertheless, it organizes the pedagogical material in the form of rules that would predominate in keyboard treatises through the end of the eighteenth century. Moreover, the very first rule given is perhaps the fundamental principle of fingering throughout the centuries. Julane Rodgers in "Early Keyboard Fingering, ca. 1520-1620" summarizes this first rule:

> The finger for a given note is determined by the notes that follow and the fingers which must be available to play them . . . one must not place at random any finger on any key, but must use the finger which would best serve in the sequence which follows.[9]

Elias Nikolaus Ammerbach (c. 1530-1579?) published his *Orgel oder Instrument Tabulatur* for the first time in Leipzig in 1571; he later revised it by changing the exercises and published it again in Nürnberg in 1583.[10] Both books were essentially anthologies of Lutheran chorale tunes, dances and transcriptions of various pieces. The 1571 version constituted the first printed German organ music; of major importance was its innovative notation consisting entirely of letters, which became the standard for German organ tabulature notation.

In the work's preface there are rules on fingering, followed by a number of exercises. The tutor is quite barren compared with Buchner's *Fundamentum*. There are more examples and less text with instructions, since Ammerbach found that the complexity of fingering made it inappropriate for lengthy verbal explanations.

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[8] Ibid., 17

[9] Julane Rodgers, "Early Keyboard Fingering, ca. 1520-1620" (D.M.A. diss., University of Oregon, 1971), 25.

[10] Mark Lindley, *Ars Ludendi: Early German Fingerings c.1525-c.1625* (Neuhof: Tre Fontane, 1993), 10.

> [But because] all fingering of the application cannot be explained through rules, I want to represent the same by examples through which one can easily judge in another way and manner.[11]

Ammerbach's basic philosophy of fingering follows the same guidelines as Bruchner's. He uses the same numbering system for the fingers, except the thumb, which he labels as "0," thereby suggesting its lesser importance. While he recommends the use of second and fourth (in modern fingering system) for the metrically accented notes, there is a significant difference from Buchner's philosophy: Ammerbach suggests the use of the left thumb even where it falls on a B-flat.

Rodgers observes that:

> Buchner's left hand fingerings are a mirror inversion of the fingerings he gives for the right hand. . . . About forty or fifty years later, Ammerbach begins to recognize finger usage peculiar to each hand.[12]

Ammerbach's explanations may not be thorough, but his examples set the foundation of the philosophy of fingering exercises that is essential even for modern keyboardists. The exercises, grouped into a figure of four consecutive stepwise notes, are extraordinarily similar to those of Hanon's infamous *Le pianist virtuose.*[13]

An additional German source of the period is a ricercar by Christian Erbach (c.1570-1635) that is preserved with fingerings.[14] The work dates from around 1625 and its suggested fingerings follow the aforementioned tutors with the use of the second and fourth fingers on the beat, and the rare and exclusively left-hand usage of the thumb.

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[11] Elias Nicolaus Ammerbach, *Orgel oder Instrument Tabulatur.* (Leipzig: Jacob Berwalds Erben, 1571), microfilm from British Museum; trans. Julane Rodgers in "Early Keyboard Fingering, ca. 1520-1620," 189.

[12] Julane Rodgers, 26.

[13] Mark Lindley, *Ars Ludendi: Early German Fingerings c.1525-c.1625*, 19.

[14] Ibid., 12.

Spanish sources

The Spanish tutors are among the most comprehensive sources for keyboard performance practice during the sixteenth century. Juan Bermudo's (1510-1559?) *Declaracion de instrumentos musicales* dates back to 1555. It is a very ambitious work consisting of five books with the announcement of two additional ones that were never actually published.[15] The keyboard fingering instructions are in the fourth book.

Bermudo employs the usual method of rules with examples, mentioning a number of exceptions that may be applied to each rule. He also provides a four-voice piece, but urges the student to apply the fingering following the preceding rules. He uses the modern way of naming the fingers and certainly makes use of the thumb for both hands. For example, the fingering for the right hand ascending scale is 1234-1234. Nevertheless, there is no description of a pivoting role for the thumb, and the little finger is essentially unused.

Despite its occasional lack of clarity, especially because of the lack of any large-scale fingered examples, the treatise includes a comment that links directly to a later approach in keyboard instruction:

> You must exercise and facilitate all the fingers, because such [a] passage can come [along in] which you may have need of all of them.[16]

In 1557 Luys Venegas de Henestrosa (c.1510-1570) compiled the first collection of Spanish keyboard music to be printed in Spain, naming it *Libro de cifra nueva para tecla harpa y vihuela*.[17] This collection is not an instruction book; therefore it presents no significant organization. However, the introduction provides advice for the performance of the pieces, and some fingering information is included.

Henestrosa numbers the fingers similarly to Bermudo, and starts the right-hand scale passages with the thumb; but unlike Bermudo's approach, the continuation of the scale involves the alternation of the fourth and third fingers. For example, the right hand

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[15] Julane Rodgers, 35-36.

[16] Bermudo, Juan. *Declaracion de inst[r]umentos musicales*. [Ossuma: Juan de Leon], 1955. Microfilm from the Library of Congress; trans. Julane Rodgers in "Early Keyboard Fingering, ca. 1520-1620," 203.

[17] Barbara Sachs and Barry Ife, ed. and translated, *Anthology of Early Keyboard Methods* (Cambridge: Gamut Publications, 1981), 68.

ascending scales would be 1234 3434 (the left hand ascending would be 4321 2121). Since it is not a comprehensive manual, there is some ambiguity concerning fingering instruction. Nevertheless, it is the first written reference in history for specifically crossing the third finger over the thumb in a descending scale that is fingered: 54321 321 321.[18]

Tomás de Santa María (?-1570) included the most detailed and comprehensive keyboard fingering instruction of the sixteenth century in his treatise *Arte de tañer fantasia, assi para tecla, como par vihuela, y todo instrumento*. The work, commonly referred to as *Arte,* dates from ca. 1541-1557; but due to a shortage of paper the treatise was not printed until 1557.[19] The *Arte* is divided into two books, with the first one including a section on keyboard technique.

Not only does Tomás de Santa María provide detailed descriptions of arm and hand position—a feature not uncommon in other treatises of the time—but he discusses all finger motions elaborately, such as the exact angle that the fingers need to bend in order to achieve overlapping, and the part of the key they need to strike. He promotes the idea of strong and weak fingers:

> It should be noted that the right hand has one principal finger and the left hand two. That of the right hand is the third finger, which is the middle one, and the two of the left hand are second and third.[20]

Arte's main contribution to the understanding of fingering of the time is the concept of giving alternative fingerings for different note values. [21] In addition, the treatise includes detailed fingerings for ornaments, intervals and even the short octave in the bass.[22]

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[18] Julane Rodgers, 46.

[19] Ibid., 49.

[20] Tomás de Santa Maria. *Arte de tañer fantasia, assi para tecla como par vihuela, y todo instrumento*, Valladolid: F. Fernandez de Cordoua, 1565, microfilm from the Library of Congress; trans. Julane Rodgers in "Early Keyboard Fingering, ca. 1520-1620," 229.

[21] Barbara Sachs and Barry Ife, 8.

[22] The short octave was an early keyboard device aimed to extend the lowest octave of the instrument by omitting some of the chromatic notes, since the bass part of the keyboard repertoire was predominantly diatonic. In this respect the lowest notes of the keyboard were tuned to pitches below their apparent ones. For example, in the case of the C/E short octave the keys which would normally be E-F-F#-G-G# were tuned as C-F-D-G-E. Source: Nicolas Meeùs, "Short Octave," *The New Grove Dictionary of Music Online,* ed. L. Macy Accessed [10/21/04], <http://www.grovemusic.com>

The in-depth theoretical analysis of keyboard instruction issues is brightened with practical suggestions described in a casual manner, undoubtedly reflecting Tomás de Santa María's own teaching experience:

> The hands [must] be placed hooked, like the paws of a cat, in such a manner that between the hand and the fingers there will in no way be any curvature; instead the knuckles have to be very sunken, in such manner that the fingers are higher than the hand [and] arched.[23]

Antonio de Cabezón (ca. 1500-1566) composed *Obras de musica para tecla, apra y vihuela,* a work that included pieces arranged according to difficulty. However, it was not published until twelve years after his death, in 1578, along with an introduction written by Cabezón's son, Hernando.[24]

This introduction gives only general information about fingering and reflects the performance practices of Hernando's time. He suggests the use of paired fingerings for the right hand and predominantly consecutive fingers for the left.[25]

Finally, Franciso Correa de Arauxo (c. 1576-1663) wrote in 1626 the treatise *Libro de tientos y discursos de musica practica, y theoritica de organo intitulado facultad organica.*[26] He provides a number of rules and examples, suggesting typically paired fingering, even though examples with three-note and four-note fingered groupings exist as well. In general, Spanish sources are the first ones to use the thumb extensively, especially for the left hand, and the first ones to number the fingers in the modern fingering system.

<u>English sources</u>

The English virginal school reached its zenith between 1575 and 1625 with the works of William Byrd (1543-1623), John Bull (ca. 1562-1628), and Orlando Gibbons

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[23] Tomás de Santa Maria; trans. Julane Rodgers in "Early Keyboard Fingering, ca. 1520-1620," 219.

[24] Julane Rodgers, 83.

[25] Barbara Sachs and Barry Ife, 66.

[26] Newman Wilson Powell, 38-39.

(1583-1625).[27] Despite the number of compositions from this time, there are no theoretical English treatises on virginal playing, nor is there any information on the introduction of the manuscripts or the published music of the time.

There is, however, a great deal of fingering in most virginal books. It is practically impossible to determine exactly when fingerings were added to the compositions, but their remarkable consistency permits the extraction of some basic principles of English fingering between 1550 and 1650.[28]

The following sources of virginal music are at least partially fingered: *My Ladye Nevells Booke; Clement Matchett's Virginal Book* (1612)*; British Museum, Add. Ms 30485* (ca. 1590-1610)*; Paris Conservatoire, Res. 1185; Fitzwilliam Virginal Book; Benjamin Cosyn's Virginal Book* (1606-1620)*; Will Foster's Virginal Book; Parthenia In- violata or Mayden-Musicke for the Virginalls and Bass Vio* (1611)*; Christ Church, Oxford, Music Ms 431* 9 (ca. 1625)*; British Museum, Add, Ms 36661* (1630); and *Priscilla Bunbury's Virginal Book.*[29]

This system of numbering the fingers is essentially the same as the modern fingering system for the right hand, but is reversed for the left hand: the little finger is numbered "1" and the thumb is "5". English sources are faithful to the idea of paired fingerings, like other sources of the time; but unlike German, Spanish or Italian sources, they favor the use of the third finger for the right hand and the third finger and thumb for the left hand on the pulse notes. For instance, a typical fingering of a right hand ascending scalar passage is 34 34 34, while the descending for the same hand would be fingered 32 32 32. For the left hand, the ascending and descending scales would be fingered 321 21 21 and 34 34 34 (in modern fingering) respectively. These are certainly not rules without exceptions, but there is an overall tendency to reserve the second and fourth fingers of both hands for the notes on "weak" beats.

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[27] Willi Apel, *The History of Keyboard Music to 1700,* trans. and rev. by Hans Tischler (Bloomington: Indiana University Press, 1972), 293.

[28] Cynthia Qualls Ashley, 5.

[29] Julane Rodgers, 119-123.

Italian sources

Girolamo Diruta's (c. 1560-?) *Il Transilvano dialogo sopra il vero modo di sonar organi e instrumenti da penna* is the earliest Italian tutor on keyboard performance. It is published in two parts. The first dates from 1597 and discusses notation, scales, and technical aspects of keyboard playing such as fingering and ornamentation; a set of toccatas on twelve church tones concludes this part. The second part, dating from 1609, includes transcriptions of various vocal compositions into keyboard versions, and discusses diminution, counterpoint and transposition.[30]

The treatise is a dialogue, an unusual format for keyboard tutors which typically demonstrate a predilection towards rule-based organization. However, the format does not minimize the comprehensiveness of the treatise. In the *Anthology of Early Keyboard Methods* Barbara Sachs and Barry Ife note:

> Il Transilvano is very complete: it tells the aspiring organist how to play, practice, finger, embellish, transpose, accompany and combine registers. It gives a rule for solmization, teaches the technique of intabulation, strict and ordinary counterpoint, and explains the quality of the modes (tuoni). In addition it contains 13 toccatas, 2 canzonas, 13 ricercars by twelve composers, liturgical settings and canti firmi.[31]

Diruta distinguishes the notes and fingers as "good" ("*buono*") and "bad" ("*cattivo*"). He teaches the use of a good finger on a good note (essentially a note in a metrically strong position) and the bad finger on a bad note (a note in a metrically weak position).[32] The numbering of fingers is identical to the modern method and the designated good fingers are the second and fourth, while the bad fingers are the first, third and fifth for both hands. The paradox of the characterization of a finger as "bad" was even mentioned by Diruta himself.

> Diruta remarked, in *Il Transilvano* in 1597, that because the third finger "must play all the bad notes, and again all the bad notes which skip," it seems to be "the

[30] Cynthia Qualls Ashley, 28.
[31] Barbara Sachs and Barry Ife, 34.
[32] Julane Rodgers, 98.

> hardest worked, since nothing usual is done without it." But this observation didn't stop him from regarding the finger as "bad."[33]

The dialogue form of the treatise makes it an ideal vehicle for an occasional casual approach that directly demonstrates Diruta's teachings:

> Above all, you must recall in what manner you have to hold the hand level with the arm, how it must be somewhat cup-shaped and the fingers curved and evenly positioned so that one is not higher than the other.[34]

One of the most remarkable contributions of the treatise, which is not related to fingering teachings, is its differentiation of keyboard instruments, particularly the organ and the cembalo. Diruta came from the Northern Italian tradition with an unmistakable preference for the organ. The Italian cembalo tradition which was centered around Naples was acknowledged by Diruta, but may have been less respected by him, since he speaks of cembalists as "Players of dances" ("Sonatori de balli").[35]

The only other contemporary source of Italian fingering is Adriano Banchieri's (1567-1634) *Conclusioni nel suono dell' organo* (1608). This is in the form of a letter "to a virtuous young organist" that contains some information on hand positions and fingerings.[36] The source is certainly not comprehensive and refers mainly to intervallic fingering.

Dutch sources

The fingering practices of Jan Pieterson Sweelinck (1562-1621) and his students Heinrich Scheidemann (1596-1663) and Samuel Scheidt (1587-1654) are found in some of their existing manuscripts. These fingering indications place the third finger on metrically strong beats in the right hand and the second and fourth fingers in the left

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[33] Ruth Nurmi, *A Plain & Easy Introduction to the Harpsichord* (Albuquerque: University of New Mexico Press, 1974), 80.

[34] Girolamo Diruta, *Il Transilvano*, Venice: Alessandro Vincenti, 1625; trans. Edward John Soehnlein, in "Diruta on the Art of Keyboard Playing: An Annotated Translation and Transcription of *Il Transilvano* Part I (1593) and II (1609)," (Ph.D. diss., University of Michigan, 1975), 148.

[35] Julane Rodgers, 95-97.

[36] Mark Lindley and Glyn Jenkins, "Fingering: Keyboard," in *The New Grove Dictionary of Music and Musicians,* ed. Stanley Sadie (London: Macmillan Publishers Limited, 2001), Vol. 8, 832-833.

hand.[37] It is, in a sense, a combination of the English method for the right hand and the Italian for the left hand.

Discussion

The earliest sources on keyboard fingering, regardless of their origin, are based on the premise that the keyboardist should not treat the fingers evenly, due to their difference in length and position within the hand. The three middle fingers are used predominantly in a manner in which the longer should cross over the shorter when playing consecutive notes. Thus, a paired fingering pattern is produced in a scale setting. The thumb and the little finger are used sparingly or not at all in passage-work. On the other hand, their use is indicated or even required for the execution of passages featuring chords and octaves. This performance practice was the common keyboard technique of the period, and was related to both the instruments and the particular compositional style of the time.

Although some early keyboard instruments—especially organs—had rather wide keys, most of them had narrower keys than the modern piano. As a result, the use of paired fingerings is unnatural for the contemporary pianist, although early fingerings can be learned and comfortably executed on today's "period" instruments. Unfortunately, the earliest of keyboard tutors rarely refer to specific types of keyboard instruments; neither do they differentiate teaching methods according to particular instrumental characteristics. An awareness of keyboard technique was just starting to develop, and authors tried to summarize the principles of playing on the keyboard, probably assuming that certain adjustments would be made by both teachers and students when necessary.

The philosophy behind early keyboard fingering is not clarified by the authors of the examined tutors, because of the sparse character of the majority of the manuals. However, the study of early fingering in conjunction with the study of early keyboard music can be helpful in drawing conclusions regarding certain performance practices. Fingering alone can not prove any articulation effect, even though certain fingerings are more conducive to a particular articulation than others. Fingering also can not

[37] Cynthia Qualls Ashley, 55.

indisputably produce strict rhythmic execution or rhythmic unevenness. Nevertheless, the grouping of fingerings promoted by a paired fingering system seems to be related to the compositional technique of diminution in much of the early keyboard repertoire, and in particular the English virginal music.

Paired fingerings are based on the fundamental idea of "strong" and "weak" fingers. The "strong" fingers are considered suitable for use on the strong beats, which were also usually moments of consonance. Even though authors from different countries do not agree on which fingers are stronger, there are similarities between the German and Italian sources that consider the second and the fourth fingers more important, and the Spanish and English sources that favor the use of the third finger in both hands and the thumb in the left hand.[38]

Despite the differences in approach, what is common to each of these early sources is the universal concern for the importance of a solid technical background for young keyboardists. The need for some systematic organization of the practical knowledge accumulated after years of teaching was evident, whether it was represented in the form of rules, dialogues, letters or even systematic fingering indications in teaching scores.

[38] Julane Rodgers, 167-168.

SOURCES FROM 1650 TO 1750

Introduction

Musicologists might argue with Claude Palisca's tracing "the beginnings of Baroque to the middle of the sixteenth century with the movement from Platonian to Aristotelian thought."[39] It is evident, however, that the Baroque followed the natural continuation of Renaissance humanism which "led to a movement in philosophy known as rationalism."[40] During the seventeenth century, philosophy and science benefited from the work of Bacon, Descartes, Leibniz, Galileo, Kepler, and Newton, to name a few.[41] However, the scientific direction certainly did not confine artistic creation to mere realistic representation. Baroque artists were deeply preoccupied with the passions of the soul and the *Affekt* of their work.

John Rupert Martin in his book *Baroque* writes:

> The seventeenth century has a Janus-like aspect: an age of extraordinary advances in philosophy and science, and of sweeping changes in the economic sphere and in the development of the modern state: but an age characterized also by continuing theological controversy, by an intense concern for the personal religious experience and by a spirit of providentialism inherited from earlier Christianity.[42]

As a result, the Baroque was "a basically new and optimistic equilibrium of religious and secular forces."[43] Hence, a large percentage of the music was still composed for the church. At the same time, the centralization of wealth and power in large centers and courts generated music production under aristocratic or royal patronage.[44]

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[39] A. Peter Brown, "Approaching Musical Classicism: Understanding Styles and Style Change in Eighteenth-Century Instrumental Music," *College Music Symposium 20/1* (Spring 1980), 7.

[40] Douglass Seaton, 151.

[41] Donald Jay Grout, 295.

[42] John Rupert Martin, *Baroque* (Boulder: Westview Press, 1977), 12.

[43] Wolfgang Stechow, "Definitions of the Baroque in the Visual Arts," *Journal and Aesthetics and Art Criticism, V*, (1946 -7), 114.

[44] David Schulenberg, *Music of the Baroque* (New York: Oxford University Press, 2001), 6-7.

The quest for reason as the chief source of knowledge did not have an immediate effect on musical treatises related to performance practices. With the exception of France, which for political reasons had an independent musical identity at the time, the number of treatises is smaller, and the documents themselves significantly less comprehensive than in the preceding century. Most sources regarding fingerings are examples of pieces fingered by composers and teachers of the time.

What is evident from almost all the sources is a gradual abandonment of the "principal" or "good" finger idea, replaced by an equal use of all the fingers, and even the pivoting role of the thumb. Perhaps the transitional nature of keyboard technique of the period did not allow enough time for theorists or teachers to absorb the new approaches and present them systematically.

German sources

Daniel Speer's (c. 1623-1693 or 1694) *Grundrichtige/Kurtz-Leicht-und Nöthiger/jetze Wol-vermehrter Unterricht der Musicalischen Kunst* was written in 1687 in a question and answer form. Though lacking detail, it promotes paired fingering with a preference for the third finger in the right and the second finger in the left hand.

Perhaps the most widely used German keyboard treatise of the second half of the seventeenth century was the *Wegweiser der Kunst die Orgel recht zu schlagen* (author unknown), which was first published in 1689.[45] The treatise contains four chapters of the *Ars cantandi* by Giacomo Carissimi[46] and continues to give lesser importance to the thumb by numbering it as "0".[47] All three middle fingers, though, seem to be placed on metrically strong notes at times, even though the preference for the third is still evident.

Johann Baptist Samber's (1654-1717) *Manuductio ad organum* of 1704 is an instruction book that includes rules of fingering. The thumb is used occasionally in both hands and is numbered both "0" and "1". In general, however, the second and fourth

[45] Cynthia Qualls Ashley, 57.
[46] Ibid., 58.
[47] Newman Wilson Powell, 123.

fingers of both hands are placed on metrically strong notes, a practice rather unusual for the German tradition at this time.[48]

German sources from the early eighteenth century gradually promote equality of all fingers. The *Anfangs-Gründe des General-Basses* by Lorentz Mizler von Kolof (1711-1778) was published in 1739 and, though it discusses primarily figured bass principles, it is the first German work to present a fingering system with the goal of being functional for remote keys.[49]

Franz Anton Maichelbeck's (1702-1750) *Die auf dem Clavier lehrende Caecilia* from 1738 is the last to assign the number "0" to the thumb.[50] Nevertheless, the thumb is widely used since all scale patterns are divided into groups of four, omitting only the little finger.[51]

Finally, special consideration needs to be given to the three fingered pieces by Johann Sebastian Bach (1685-1750), not only because of his indisputable contribution to keyboard literature, but also because they demonstrate the transition to a new fingering approach. *Applicatio* (BWV 994) and *Praeambulum* (BWV 930) are preserved in Bach's handwriting, while *Prelude and Fugetta* [sic] (BWV 870a) has survived in a manuscript whose scribe has been identified "as Johann Caspar Vogler, Bach's student and successor at Weimar."[52] The fingering in these sources suggests that Bach used the older paired fingerings for predominantly white-note keys and adopted modern fingerings for the remote keys.[53] The pedagogical value of the sources is highlighted by the fact that the first two pieces were used in a teaching collection compiled by Bach for his son Wilhelm Friedemann.

Carl Philipp Emanuel Bach makes reference to his father's fingering approach in his treatise *Versuch über die wahre Art das Clavier zu spielen*:

> My deceased father told me that in his youth he used to hear great men who employed their thumbs only when large stretches made it necessary. Because he lived at a time when a gradual but striking change in musical taste

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[48] Ibid., 148.
[49] Cynthia Qualls Ashley, 60.
[50] Newman Wilson Powell, 155.
[51] Ibid., 155.
[52] Quentin Faulkner, *J.S. Bach's Keyboard Technique: A Historical Introduction* (St. Louis: Concordia Publishing House, 1984), 13.
[53] Mark Lindley, "Keyboard Technique and Articulation," in *Bach, Handel, Scarlatti: Tercentenary Essays,* ed. Peter Williams (Cambridge: Cambridge University Press, 1985), 228.

> was taking place, he was obliged to devise a far more comprehensive fingering and especially to enlarge the role of the thumbs and use them as nature intended.[54]

Italian sources

Lorenzo Penna's (1613-1693) *Li primi albori musicali per li principianti della musica figurata* of 1684 is the only Italian treatise of this period.[55] The tutor consists of three books whose main concern is the teaching of thoroughbass and counterpoint, but the last book contains a brief fingering section in the usual rule format. From the given rules and examples it is evident that the fingering system described by Diruta is at this time not the only one in use, since here the third finger plays all the metrically important notes, rather than the second and the fourth fingers prescribed by Diruta.

A significant source of Italian fingering is the fingered *Toccata primo* by Alessandro Scarlatti (1660-1726). Scarlatti uses an imaginative fingering system with a symbol representing each finger. These symbols correspond with modern fingering as follows:

*	|	∝	◜	▲
1	2	3	4	5

The *Toccata* is fingered throughout in a manner equivalent to modern scale fingering, with ample use of all five fingers.[56] The occurrence of some ascending right hand passages and descending left hand passages, where the fingering as described by Diruta is still in use, proves the transitional nature of Alessandro Scarlatti's keyboard technique.[57]

Domenico Scarlatti provided no direct evidence or teaching of keyboard fingerings. Nevertheless, the complexity of the keyboard writing as evidenced in his

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[54] C.P.E. Bach, *Essay on the True Art of Playing Keyboard Instrument*s, trans. and ed. by William J. Mitchell (New York: W.W. Norton & Company, 1949), 42.

[55] Newman Wilson Powell, 122.

[56] Ralph Kirkpatrick, *Domenico Scarlatti* (New Jersey: Princeton University Press, 1953), 188.

[57] Mark Lindley, "Keyboard Technique and Articulation," 213.

Essercizi—replete with rapid successions of double thirds, hand crossings, and extended scale passages—suggests a fingering approach that is close to the modern conception.[58]

Ralph Kirkpatrick describes Domenico's fingering:

> Like J.S. Bach and Rameau, Scarlatti must have early cultivated a system tending toward equal development and independence of the five fingers of each hand. . . . It is probable that, like C.P.E. Bach, Scarlatti retained the old fingerings for certain passages and made use in others of the modern principle of passing the thumb under in scale passages.[59]

English sources

The first form of fingering instruction from England is the preface to Henry Purcell's (c.1659-1695) *A Choice Collection of Lessons for the Harpsichord or Spinet*[60] from 1696. This introduction contains an example of a fully fingered C major scale. The numbering of the fingers from "1" through "5" moves from left to right for both hands, the left-hand numbering thus mirroring the modern way.

In "Phrasing and Articulation in Henry Purcell's Harpsichord Suites" Carey Diane Bozovich comments on the fingering instructions of this introduction:

> The directions at the side of the written-out scale state: "Right hand [sic] the Fingers to ascend are the third & fourth to decend [sic] ye third & second"; "Left hand [sic] the fingers to ascend are ye third & fourth to decend [sic] ye third & second.[61]

While it is evident from the above suggestion that Purcell is influenced by the paired fingering of the Virginal School, the anonymous source entitled *The Harpsichord Master,* first published in 1697 by I. Walsh, includes a "Prelude for Ye Fingering by Mr. H. Purcell." This source uses the same initial fingered scale as Purcell's, reflecting the influence of the principles of the Virginal school well into the eighteenth century. Both sources are valuable for that information, but the lack of any sufficient explanatory

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[58] Richard Boulanger, *Les Innovations de Domenico Scarlatti dans la technique du clavier* (Beziers: Société de musicology de Languedoc, 1988), 213.

[59] Ralph Kirkpatrick, 188.

[60] Henry Purcell, *Works for Harpsichord and Organ* (New York: Lea Pocket Scores, 1968), V.

[61] Carey Diane Bozovich, "Phrasing and Articulation in Henry Purcell's Harpsichord Suites," (Master's Thesis, Andrews University, 1985), 84-86.

material underlines a long English tradition according to which the keyboard teaching is based solely on musical material rather than on explanatory tutors.

Peter Prelleur's *The Harpsichord Illustrated and Improv'd [sic]; Wherein is Shewn the Italian Manner of Fingering with Suits of Lessons for Beginners & Those who are Already Proficients [sic] on the Instrument and the Organ* certainly marks a shift in this tradition. This instruction book is part of a volume that includes tutors for various instruments bound together; it was published in 1731. [62]

The short fingering section of the tutor is aimed at training the aspiring keyboardist. The first examples are fully fingered; gradually the fingering suggestions are given more sparingly, until the student is able to decide without any hint what the proper fingering should be. The numbering is identical to the modern system and the thumb is used extensively in both hands; the little finger is avoided.

At the beginning of the fingering section by Prelleur himself, the spirit of separation from inflexible rules is well described:

> Although there is no certain rule to be laid down for fingering of any Tune that you may meet with yet the following Lessons may be a great Inlet to it if well observed.[63]

<u>French sources</u>

France continued its independent musical development throughout the Baroque period. In *Baroque Music* Claude Palisca explains:

> French musicians in the seventeenth century enjoyed a long period of relatively undisturbed cultivation of their own soil. Even instrumental idioms—which lacking a choice tie to language habits, tend to be homogenized by border-crossing musicians, printed editions and instruments—preserved in France a distinct character.[64]

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[62] Marvin John Bostrom, "Keyboard Instruction Books of the Eighteenth Century" (Ph.D. diss., University of Michigan, 1960), 29.

[63] Peter Prelleur, *The Harpsichord Illustrated and Improved, wherein is Shewn the Italian Manner of Fingering*. (London: Printing Office in Bow Church, 1731), 5.

[64] Claude Palisca, *Baroque Music* (New Jersey: Prentice Hall, 1968), 175-176.

The main reasons for this musical isolation are the strong nationalistic spirit cultivated by King Louis XIV after the Thirty Years War (1618-1648) and France's success in asserting its integrity against the Hapsburg Holy Roman Empire.[65] The arts were controlled by official policy, and the strong guild of musicians did not allow the development of independent musical trends.

During the seventeenth century, the primary French sources available to harpsichordists were ornament tables appearing in collections of pieces by various French composers, such as Jacques Champion de Chambonnières and Jean-Henry D'Anglebert. There are, however, two organ sources that include fingering suggestions: the *Livre d'orgue* of 1665 of Guillaume-Gabriel Nivers (1632-1714), and the *Livre d'orgue* of 1688 by André Raison (ca. 1640-1719).[66] Nivers' guide includes some fingering suggestions that promote paired fingering, using the third and fourth fingers for ascending right hand scales, and the third and second for descending right hand scales, while the fingerings for the left hand ascending and descending scales were 21 21 and 34 34 respectively. Raison's book, on the other hand, does not correlate fingering with metrically strong notes, though it does make use of the little finger.

The first French instruction book on harpsichord playing is Monsieur[67] de Saint Lambert's (c. 1700) *Principes du clavecin* of 1702.[68] The tutor has significant pedagogical value due to its extensive details in all elements related to harpsichord playing, such as clef reading, notes inégales, ornamentation, tempi and so forth. The treatise includes twenty-eight chapters, of which the nineteenth is devoted to fingering.

In his assessment of the book, Bostrom concludes:

> The Saint Lambert method is "clear and comprehensible" to the untutored; it is accurate; the order is logical; it is explicit and self-explanatory. The Saint Lambert method sets the pattern for those that follow.[69]

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[65] Douglass Seaton, 183.

[66] Cynthia Qualls Ashley, 69-70.

[67] The common perception that Saint Lambert's first name was "Michel" derives from the confusion of Saint Lambert and the singer/composer Michel Lambert, an error that goes back at least as far as Walther's *Musicalisches Lexicon* (1732), according to Rebecca Harris-Warrick, "Saint Lambert" in *The New Grove Dictionary of Music and Musicians,* ed. Stanley Sadie (London: Macmillan Publishers Limited, 2001), Vol. 22, 102-103.

[68] Rebecca Harris-Warrick, 102-103.

[69] Marvin John Bostrom , 26.

The suggested fingering for scales is identical to that given by Nivers. Nevertheless, Saint Lambert's opening paragraph allows certain latitude for the strictness of any given rules.

> There is nothing more free in harpsichord playing than the position of the fingers. Everybody seeks the most convenient and favorable way. But there are situations where all those who play use their fingers the same way and because it is recognized that this is the most suitable thing to do. For this has become established as a sort of rule which we feel we are almost obliged to follow and which a beginner especially ought not neglect.[70]

In the preface of the treatise, Saint Lambert expresses his wish to be precise and scientific. His fingering suggestions are indeed specific, as he gives even alternative intervallic fingering for large and small hands. The scientific background of his suggestions is also evident:

> With regard to the aptitude of the hands, there is no one who cannot have it if he begins to exercise early. Since that aptitude is nothing other than a great suppleness in the nerves which permits the fingers the liberty of moving artfully, childhood is the most proper time to develop it.[71]

François Couperin's (1668-1733) celebrated *L' art de toucher le clavecin* from 1716 is not as thorough and organized as Saint Lambert's treatise; nevertheless, fingering is a primary focus. The different philosophies of the two treatises are evident from the authors' statements of purpose. While Saint Lambert strives to make the "art understandable from his book without the help of anyone,"[72] Couperin provides principles that are "absolutely necessary to succeed in playing [his] pieces well."[73]

L' art de toucher le clavecin is loosely organized in narrative style, including discussion on performance practice issues such as ornamentation and fingering. In

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[70] Monsieur Saint Lambert *Principes du clavecin*; trans. in William Neil Roberts, "The Harpsichord Instruction Books of Michel de Saint-Lambert and François Couperin: A Discussion of their Content and Comparative Description of their Agréments" (Master's thesis, University of Washington, 1962), 22.

[71] Ibid., 12.

[72] William Neil Roberts, "The Harpsichord Instruction Books of Michel de Saint-Lambert and François Couperin: A Discussion of their Content and Comparative Description of their Agréments," 9.

[73] François Couperin, *L'art de toucher le clavecin,* trans. and ed. by Margery Halford (Van Nuys: Alfred Publishing Company, 1995), 28.

addition, it includes an *Allemande* and eight *Préludes*. Couperin devotes large portions of his treatise to fingering, believing that "the manner of fingering does much for good playing."[74]

In the given examples, a combination of paired and modern fingering is used. The thumb and the fifth finger are used fairly freely, and paired fingering, when used, is not directly associated with metrically strong notes.[75] Additionally, alternate fingerings for playing consecutive thirds are given, and there are numerous suggestions for finger substitution to allow longer and more legato lines. The growing consciousness that fingers should be treated or at least trained as equal is also reflected in Couperin's reference to fingering suitable for trills.

> Many people have less aptitude for playing trills and appoggiaturas with certain fingers: in these cases, I advise them not to neglect to try to improve them by many exercises. But, at the same time, as the better fingers become more perfect, they should be used in preference to the weaker ones without any regard for the old style of fingering, which must be given up in favor of the good playing expected today.[76]

The last French treatise that provides information on French Baroque keyboard practice is the *Méthode de la méchanique des doigt sur le clavecin* by Jean Philippe Rameau (1683-1764). This treatise appeared as a preface in the 1724 publication of Rameau's *Pièces de clavecin*.[77] The main focus of the material, which is in essay style, is the description of finger action. There is no direct reference to fingering choice, but Rameau also includes a fully fingered piece as an example, the *Menuet en Rondeau.*

From this piece, it is evident that Rameau's fingering is based on a modern concept with ample use of the thumb and the little finger. In the preface he also mentions the influence that these two fingers have on the overall hand position:

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[74] Ibid, 31.

[75] Sandra Soderlund, *Organ Technique: An Historical Approach* (Chapel Hill: Hinshaw Music, 1982), 108.

[76] François Couperin, 32.

[77] Cynthia Qualls Ashley, 81.

> When the thumb and the little finger . . . rest on the edge of the keys, they make it necessary for the other fingers to curve, so that these, too, may rest on the edge of the keys.[78]

Discussion

While most treatises from the second half of the seventeenth century seem to promote paired fingerings, there is certainly a noticeable change in the last years of the seventeenth and the first half of the eighteenth century. Composers and teachers from different regions seem to recognize gradually the importance of using the thumb and the little finger.

Nevertheless, this technical change is not discussed thoroughly in the treatises. Most keyboard tutors are written in a sparse manner, providing only few rules and examples of fingering. In fact, even the number of treatises from this particular time is reduced compared with the first generation of keyboard tutors. Perhaps the major change of keyboard technique that was taking place due to the increasing use of the thumb and the little finger—which coincided with the compositional changes of the late Baroque—created a certain hesitation to standardize keyboard practices.

In addition, the regions of Europe that generated the majority of publications on keyboard performance practice changed. Spanish writers who provided the most detailed treatises in the past did not contribute in this era of transition; English tutors appeared for the first time but were still extremely limited; and Italian sources were mainly based on pieces fingered by composers.

The most significant and comprehensive sources of this period come from Germany and France. French sources, in particular, acknowledge the increasing need for appropriate keyboard instruction. The willingness of the middle class to acquire a musical education was not always rewarded with effective keyboard teaching, thus the need for publications with organized keyboard teaching material was stronger than ever.

As Saint Lambert explains:

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[78] Jean-Philippe Rameau, *Piéces de clavecin*, trans. Erwin R. Jacobi (London: Bärenreiter Kassel, 1966), 18.

> I know of no more considerable fault in a harpsichord master than that of not knowing how to place the hand of his student and of showing them bad usage of their fingers. . . . Since this fault always comes from the master who first taught us, it is important to choose one who knows how to avoid it.[79]

[79] Monsieur Saint Lambert, 15-16.

CHAPTER 2

CLASSICAL PERIOD

SOURCES FROM 1750 TO 1800

Introduction

The cultural movement that prevailed in Western Europe during the second half of the eighteenth century was the Enlightenment. This movement which had its roots in English empiricism, French rationalism and French skepticism promoted the ideals of clarity and formal symmetry, as well as the ideal of education as one of society's primary goals.[80] The scientific achievements of this particular time, such as the revolutionary theory of chemical elements by Lavoisier and the countless machines that were developed as a result of the innovation of the steam engine by Thomas Newcomen, provided practical changes beyond the idealistic quests.

The rise of the bourgeoisie resulted in a powerful and wealthy middle class, with an immense interest in participating as spectator or even as amateur performer in all cultural events. In the field of music, this resulted in increasing numbers of large theaters that were built to accommodate the new concept of the public concert, and in the flourishing of the music publication enterprise.[81] Consequently, composers demonstrated an inclination toward simplified means in order to appeal to the amateurs, and the performance practice tutors became more explicit.

Charles Rose in *The Classical Style* writes:

> Is the amateur nature of most keyboard music of the latter half of the eighteenth century due to the fact that the pianoforte became the particular province of the female musician? Most of Haydn's piano sonatas and piano trios, many of Mozart's concertos and Beethoven sonatas were especially written for ladies.[82]

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[80] Daniel Hertz, "Enlightenment" *The New Grove Dictionary of Music Online*, ed. L. Macy (Accessed [6/7/04]), 1.

[81] Douglass Seaton, 238.

[82] Charles Rosen, *The Classical Style*, (New York: W.W. Norton & Company, 1997), 46.

The keyboard treatises of that time reached a new level of organization and precision, partially because of the amateur nature of their targeted audience, and partially because of their aspiration to gather all the knowledge available in an encyclopedic manner.[83] As a result, most treatises focus on matters of musical technique rather than aesthetics alone.[84] In addition, they reflect a concern that has been observed already in the last treatises of the first part of the century: the inadequacy of the keyboard teacher.

Bostrom writes:

> The keyboard treatises published during the eighteenth century seem to have been written as much for the edification of teachers of keyboard instruments as for students. . . . Saint-Lambert went on to give considerable additional space to the attributes of a good teacher. . . . Late in the century Türk indicated that most of his remarks were as pertinent for teachers as for students.[85]

German-speaking regions seemed to be the pioneers of keyboard music of this period since they were the birthplaces of both the leading composers and the leading music theorists of the Classical era. Friedrich Wilhelm Marpurg, Carl Philipp Emanuel Bach, and Daniel Gottlieb Türk created the prototypes for modern keyboard instruction, influencing generations of future keyboardists.

Carl Philipp Emanuel Bach (1714-1788): *Versuch über die wahre Art das Clavier zu spielen*

C.P.E. Bach's 1753 treatise on keyboard playing is unquestionably one of the most important books of its kind. Its organization, thoroughness, and in particular its reflection on the technical demands that music of its time required, assured an unprecedented acceptance throughout Europe. At a time when music publications sold only a few dozen copies, *Versuch* allegedly sold close to fifteen hundred copies before the end of the eighteenth century.[86]

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[83] Philip G. Downs, *Classical Music: The Era of Haydn, Mozart and Beethoven* (New York: W.W. Norton & Company, 1992), 9.

[84] Ibid., 12.

[85] Marvin John Bostrom, 9.

[86] Roger Crager Boardman, "A History of Theories of Teaching Piano Technic" (Ph.D. diss., New York University, 1954), 15.

Roger Crager Boardman, in "A History of Theories of Teaching Piano Technic," analyzes its vast influence on the music world:

> That both Bach and the *Essay* were held in high esteem is revealed in the endorsements of several of the recognized men of music. Clementi stated that he owed all his knowledge and ability, his new touch, his fingering and new style to this book alone. Haydn called it the "school of schools." Beethoven used this book in teaching and closely followed the *Essay* in instructing the young Czerny.[87]

The treatise is written in two parts, the second of which is mainly dedicated to thoroughbass principles and is organized into chapters. The fingering section is quite extended and has a prominent role as the first chapter of Part One. C.P.E. Bach discusses the subject thoroughly with an abundance of examples. Not only does he emphasize hand position, he also connects the shape of the keyboard with the anatomy of the human hand in order to validate his fingering choices.

> The shapes of our hand and the keyboard teach us how to use our fingers. The former tells us that the three interior fingers are longer than the little finger and the thumb. From the latter we learn that certain keys are longer and lie lower than the others . . . the black keys belong essentially to the three longest fingers. Hence, the first principal rule: Black keys are seldom taken by the little finger and only out of necessity by the thumb.[88]

His professional and systematic approach to the matter is also evident in the elaboration on the fingerings of each scale. C.P.E. Bach provides basically all of the standardized fingering that we use even today but, acknowledging also the individuality of each hand, he provides alternative fingering for some scales. Perhaps his biggest theoretical contribution is the clear reference to the pivotal role of the thumb.

> Our five fingers can strike only five successive tones, but there are two principal means whereby we can extend their range as much as required, both above and below. They are the turning of the thumb and the crossing of the fingers. . . . Of the five fingers, the thumb alone is naturally adept at turning under. Flexible and propitiously short, it is the only one to be concerned with this technique, which is

[87] Ibid., 16.
[88] C.P.E. Bach, 45.

> employed when the fingers, playing in the normal order, cannot compass the range of a passage.[89]

C.P.E. Bach mentions the older fingering, and occasionally includes suggestions that derive from old fingering techniques; nevertheless he is very critical of it since it may be the cause of "pupils whose fingers stumble, miss and interlock."[90] He is also very critical of Couperin's overuse of the finger substitution technique, even though he considers Couperin's treatise to be otherwise "sound."[91]

There is no doubt about the importance of the fingering issue in C.P.E. Bach's perception of requirements for an adequate keyboard technique. Even though in most treatises of the Baroque era the fingering section usually is limited to a few pages in the form of rules that rarely ever exceed ten in number, *Versuch* not only devotes several pages to the topic, but it does so in the form of rules or suggestions that reach the number ninety-nine.

C.P.E. Bach states the purpose of his elaborate teaching from the beginning of the chapter:

> It can be seen that correct employment of the fingers is inseparably related to the whole art of performance. More is lost through poor fingering than can be replaced by all conceivable artistry and good taste.[92]

Friedrich Marpurg (1718-1795): *Die Kunst das Clavier zu spielen* and *Anleitung zum Clavierspielen*

Friedrich Marpurg first published his *Die Kunst das Clavier zu spielen* in 1750, although he revised it after C.P.E. Bach published the *Versuch*, and published it again in 1755. In his preface Marpurg mentions consulting the work of other authors such as Couperin and C.P.E. Bach.[93] In 1755, Marpurg also published the *Anleitung zum Clavierspielen*, which he himself translated into French and published as *Principes du*

[89] Ibid., 45-46.
[90] Ibid., 69.
[91] Ibid., 72.
[92] Ibid., 41.
[93] Marvin John Bostrom, 30.

clavecin in 1756.[94] The two works are Marpurg's main didactic keyboard treatises. The number of publications that these two works received under different titles has created a certain degree of confusion regarding the separate identity of the two books.

Elizabeth Loretta Hays in "F.W. Marpurg's *Anleitung zum Clavierspielen* (Berlin, 1755) and *Principes du clavecin* (Berlin, 1756): Translation and Commentary" elaborates on this problem:

> Together the *Kunst das Clavier* and the *Anleitung* went through no less than nine subsequent editions and translations (including the addition of a second part (1761) to the *Kunst das Clavier* of 1750-55 and two subsequent translations of that Second Part). . . . Because of the great similarity of title and content among all of these publications, most bio-bibliographies have confused all of them – including the two original works themselves – almost inextricably.[95]

Die Kunst das Clavier zu spielen in its original edition listed various fingered patterns in tables without any examples.[96] One of the revisions that Marpurg made for the 1755 edition was to add fingered examples of musical patterns. Even though the tutor has distinctive sections, an unusual element of its organization is the placement of the fingering section last, after the discussion of position, technique and interpretation. The instructions on fingering are simply fingered musical examples.

The lack of detail in the treatise impedes any analytical assertions on the pedagogy of fingering. Marpurg uses predominantly the modern system of fingering, even though the older one is not totally abandoned. He presents more examples with fingered arpeggios and chords than does C.P.E. Bach, and similarly to the latter he believes that one develops facility by "learning to play musical patterns with certain correct fingering."[97]

The *Anleitung zum Clavierspielen* is much more comprehensive and explicit than *Die Kunst das Clavier zu spielen,* at least as far as the fingering section is concerned. This section follows the discussion on ornamentation, a choice that was criticized at the

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[94] Elizabeth Loretta Hays, "F.W. Marpurg's *Anleitung zum Clavierspielen* (Berlin, 1755) and *Principes du clavecin* (Berlin, 1756): Translation and Commentary" (Ph.D. diss., Stanford University, 1976), T-v.

[95] Ibid., 58.

[96] Roger Crager Boardman, 30.

[97] Ibid., 31.

time of its publication.[98] The format of the fingering discussion is very clear and understandable, including an introduction, a set of fingering rules with many examples, presentation of scales in all the keys in both parallel and contrary motion, as well as a set of fingered patterns aimed to serve as exercises.

Marpurg's fingering suggestions are again similar to modern ones. He makes reference to the old fingerings and allows them only if they are habitual and do not interrupt the melody.[99] However, he uses very strong language to support the equal treatment of all fingers:

> One should endeavor to make all fingers equally nimble without distinction. Neither the little finger nor the thumb should be excepted . . . [One] may surely believe that teachers who exempt either from the fingering misguide those who are entrusted to them. If one had still more fingers, one could make use of all of them.[100]

While the *Kunst das Clavier* includes elementary rudiments of music, the *Anleitung* is aimed for more advanced students,[101] much like C.P.E. Bach's *Versuch*. There is a certain influence by C.P.E. Bach; nevertheless, Marpurg demonstrates a thorough knowledge of different pedagogical approaches to achieve finger facility, such as exercises in contrary motion and progressive exercises that are transposed to all the keys. He even suggests F major rather than C as a starting point because of the more comfortable hand position that F major provides.

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[98] Elizabeth Loretta Hays, 121.

[99] Friedrich Wilhelm Marpurg, *Anleitung zum Clavierspielen* (Berlin: Den Gaude und Spenner, 1765; reprint, Hildesheim: Georg Olms Verlag, 1970); trans. Elizabeth Loretta Hays in "W.F.W. Marpurg's *Anleitung zum Clavierspielen* (Berlin, 1755) and *Principes du clavecin* (Berlin, 1756): Translation and Commentary," II: 1-7.

[100] Wilhelm Marpurg, *L'art de toucher le clavecin* in Clavecin: Serie I, France 1600-1800, ed. Jean Saint-Arroman, Friderich, vol. II, Paris: Éditions Fuzeau, 2002; trans. Elizabeth Loretta Hays in F.W. Marpurg's *Anleitung zum Clavierspielen* (Berlin, 1755) and *Principes du clavecin* (Berlin, 1756): Translation and Commentary," 9.

[101] Elizabeth Loretta Hays, 89.

Daniel Gottlob Türk (1750 -1813): *Klavierschule*

Türk's *Klavierschule* was published in 1789 and is "the last major work in the tradition of C.P.E. Bach's *Essay*."[102] Like the *Essay* and Marpurg's treatises (in particular the *Anleitung*), the work aspires to provide organized information on all the elements connected to keyboard playing, as an aid to both teachers and students. Türk, in addition, attempts to appeal to readers with interest in academic research.

> This work as will be seen is intended for three classes of readers. The main text contains that which everyone, including the student, must know. The intended notes are very likely for the most part for the teacher. In the additional remarks in the footnotes, which are numbered, are found various observations which may give the researcher in music material for further thought about this or that subject.[103]

The treatise is organized in well defined chapters, of which the second is devoted to fingering. The introduction comprises a general discussion on posture, available keyboard instruments and general advice for the progression of lessons. The fingering section, in the typical "rule" approach, is placed at the beginning of the technical discussion, since the first chapter focuses on theoretical elements. Unlike all previous treatises though, *Klavierschule* organizes the fingering material into distinctive subjects.[104]

Türk's approach to fingering is stated in the introductory pages:

> All fingers must be utilized in playing, for there are certain passages which, without the thumb and the little finger, can either not be played at all or, at least, only clumsily and falteringly. . . . Our present compositions are for the most part so constituted that one often wishes for even more fingers.[105]

Similar to most writers of the second half of the eighteenth century, Türk promotes fingering that is predominantly identical to the modern approach, with explicit instruction on "putting the thumb under"[106] the other fingers, and "crossing over"[107] the

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[102] Raymond H. Haggh, "Translator's Introduction," in Daniel Gottlob Türk, *School of Clavier Playing,* translated by Raymond H. Haggh (Lincoln: University of Nebraska Press), xiii.

[103] Daniel Gottlob Türk, *School of Clavier Playing,* (1789), translated by Raymond H. Haggh (Lincoln: University of Nebraska Press), 6.

[104] Roger Crager Boardman, 39.

[105] Daniel Gottlob Türk, 31.

[106] Ibid., 133.

thumb for the rest of the fingers, with the obvious exception of the little finger. However, some latitude is allowed for the older style of fingering, demonstrating that even close to the last decade of the century the old fingering was at least partially still in use.

> One only crosses over the thumb (with one of the three longer fingers) because no finger is as short as the first, but in certain cases one can also cross over the fourth finger with a third, over the fifth finger with a fourth, and even the third finger over the little finger.[108]

Türk promotes fingering which allows the hand to remain quiet and, similar to C.P.E. Bach, is convinced of the importance of practicing specific patterns with certain fingerings. He provides fingering for all the scales, including alternative fingerings for many of them. The fingering discussion also includes an extended section on passages played by alternating hands. Furthermore, the author provides painstakingly detailed instruction and examples on playing intervals, discussing the impact of the keyboard design to formulate different hand positions for the same types of intervals.

The *Klavierschule* includes an abundance of examples which demonstrate the various rules and their exceptions. Even though the inclusion of specifically designed examples for each occasion was standard practice in all major treatises of this time, Türk also makes considerable use of music by other composers from his time or even from previous generations. Boardman notes that Türk "fostered the dissemination of good music literature by including in his text examples from the writings of J.S. Bach, Emanuel Bach, Benda, Haydn and Mozart."[109]

Other significant sources of the period

The considerable advancements in keyboard technique and the improvements in keyboard instruments with the increasing predominance of the fortepiano inspired many teachers to write manuals for keyboardists during the second half of the eighteenth

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[107] Ibid., 135.
[108] Ibid., 135.
[109] Roger Crager Boardman, 40.

century. Most of them were certainly influenced by the treatises of C.P.E. Bach, Marpurg and Türk, and as a result the majority of these later tutors lack in originality.

Georg Simon Löhlein's (1725-1781) *Klavierschule* of 1765 was the first German treatise of the time to give special attention to the beginner.[110] Constant references to bad teaching habits throughout the book express vividly Löhlein's general dissatisfaction with the keyboard teaching of his time.

The work consists of two parts, the first dealing with rudiments of music and technical issues, the second devoted to harmony and figured bass. The second volume did not appear until 1788, and by that time the first volume had already undergone additional printings.[111]

The seventh chapter of the first part is devoted to fingering, and consists of simple rules since Löhlein addresses his tutor to both teachers and students. The suggested fingering is typical of the period, that is, predominantly modern with a few reminiscent paired fingerings. Perhaps the method's main contribution is the section following the fingering discussion that includes a fully fingered collection of progressively more difficult minuets, gigues, allegros, polonaises and divertimenti.[112] Löhlein underlines the need for such a collection being designed for the benefit of a beginner:

> Because certain teachers often write nothing other than worthless pieces as beginning exercises for the students, these pieces have neither a comprehensible melody nor a system of fingering. At the onset, both the student's hearing and fingers are ruined.[113]

Special attention should also be given to Georg Friedrich Wolf's (1761-1814) *Unterricht im Klavierspielen* of 1783. This tutor, modeled after the major treatises of C.P.E. Bach and Marpurg, owes its value to its explicit explanatory remarks. According to Marvin John Bostrom, these remarks are intended "to clarify, to cite variances, to give

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[110] Dora Jean Wilson, "Georg Simon Löhlein's *Klavierschule*: Translation and Commentary" (Ph.D. diss., University of Southern California, 1979) 42-43.

[111] Ibid., 42-43.

[112] Roger Crager Boardman, 32.

[113] Georg Simon Löhlein, *Clavier-Schule* (Leipzig: Waisenhaus and Frommann, 1765); trans. Dora Jean Wilson, "Georg Simon Löhlein's *Klavierschule*: Translation and Commentary" (Ph.D. diss., University of Southern California, 1979), 125.

historical development, to point out comments by other authors, and so forth."[114] Wolf's view of the importance of fingering is nothing less than what is observed in almost all of the aforementioned treatises of the eighteenth century:

> If you do not have the right fingering you can never play a piece distinctly and fluently; instead you will stumble at each small difficulty, and how can you put expression in such playing? Ease and good demeanor are the two things on which all rules concerning this must be based.[115]

Georg Merach's *Clavierschule für Kinder* of 1789 follows Löhlein's example in addressing issues for beginners and particularly children. The chapter on fingering includes ample examples, suggestions for practice, and advice for posture, hand positions and basic principles to approach a correct method of fingering.[116]

Other books by German writers were less significant, since they do not contribute considerably to the pedagogical, theoretical or analytical approach to keyboard fingering instruction. In this category, Johann Töpfer's *Anfangsgründe zur Erlernung der Musik und insonderheit des Claviers* from 1773, Henrich Laag's *Anfangsgründe zum Clavierspielen und Generalbass* from 1774, and Johann Milchmeyer's *Die wahre Art das Pianoforte zu spielen* from 1797 should be mentioned.

The most important French treatise from the second half of the eighteenth century is Antoine Bemetzrieder's (1739-after 1808) *Leçons de clavecin et principes d'harmonie,* published in 1771. The treatise appeared in translations in English, Dutch and Spanish.[117] It is written in the form of a dialogue, which makes the separation of topics unclear. Fingering is discussed along with issues of notation, rhythm, scales and modulations in the same section. The most thorough coverage of the topic is the suggested fingering for scales which is very clear and similar to modern fingering.

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[114] Marvin John Bostrom, 44.

[115] Georg Friedrich Wolf, *Unterricht im Klavierspielen* (Halle: J.C. Hendel, 1789), 52; trans. Marvin John Bostrom, *Keyboard Instruction Books of the Eighteenth Century* (Ph.D. diss., University of Michigan, 1960), 126.

[116] Marvin John Bostrom, 43.

[117] Ibid, 41.

Bemetzrieder's goal as presented in the introduction is to help "beginners as well as those who are willing to bring their talent to perfection."[118] The format of the treatise makes it more suitable as a reference tool for teachers, since it provides answers to common student questions.[119] In addition, Bemetzrieder addresses one problem that many teachers encounter with students of very early age: the inability to teach standard fingering because of the hand size restrictions. His suggestion is logical, yet unconventional:

> Children should be first taught on Spinnets made on purpose with narrower keys than usual; . . . Thus they will be taught good fingering at first, and acquire a good habit from their earliest lessons. Otherwise they must learn their lessons with wrong fingers in their infancy, and then learn them over again with proper fingers in their riper years: which perhaps, may not be so easily done, as it is more than probable that a tincture of bad fingering will stick to them as long as they live. This last assertion I can aver by the experience I have had with some of my own scholars.[120]

The transition from paired fingerings to the modern approach, which gives a pivotal role to the thumb, is portrayed also through the large number of English treatises of this period that include information on keyboard fingering. One of the most important ones is Nicolo Pasquali's (1718-1757) *The Harpsichord Illustrated with Examples in Notes; To which is added an Approved Method of Tuning this Instrument* from 1760. This short treatise is mainly preoccupied with keyboard fingering issues. Headings are given for each particular fingering aspect, and plentiful examples illustrate the various points. In addition, fully fingered scales are provided.

Pasquali's approach to fingering reflects the inconsistencies that derive from the simultaneous use of both old and new fingering methods. He numbers the thumb as "0," implying its lesser value; nevertheless the position and use of the thumb is more often discussed than any other finger in the treatise. Pasquali at times follows the philosophy of repositioning the hand without passing the thumb under the other fingers:

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[118] Anton Bemetzrieder, *New Lessons for the Harpsichord* (London: Printed and sold by the author, 1771), 4.

[119] Marvin John Bostrom, 42.

[120] Anton Bemetzrieder, v, vi.

> We must suppose a passage consisting chiefly of consecutive or following notes exceeding the compass of five, for which two fixed positions of the hand are necessary . . . it is obvious that in the transition from one position to the other, the finger of the last note of the first position must be lifted from the key before the full time of the note be expired, in order to get the hand shifted to the first note of the second position when the exact time requires it.[121]

On the other hand, he acknowledges the requirement for uninterrupted legato playing of long passages:

> When we have not the advantage of a rest, we must have recourse to the thumb, which while it is playing will give sufficient leisure, by its shortness, to the other fingers to pass over it in descending, or whilst the other fingers are playing, it will easily pass under them in ascending.[122]

Similarly, Caspar Heck's (1740?-1791) *The Art of Fingering* advocated the "thumbs-under" technique as early as 1766.[123] However, the scale examples are fingered with the old paired-finger technique. Even twenty years later in 1785, the anonymous *Preceptor for the Pianoforte, Organ or Harpsichord*, uses the thumb extensively but still is reluctant to use the fifth finger. This tutor also functions as a workbook for fingering, since the fully fingered examples from the beginning are followed by lessons in which the fingering is left to the student's judgment.[124]

Johann Christian Bach (1735-1782) and Francesco-Pasquale Ricci (1732-1817) in their keyboard treatise *Méthode ou recueil de connoissances élémentaires pour le forte-piano ou clavecin* from 1798 strongly suggest practicing the new fingering technique:

> Become accustomed to passing the thumb under whatever finger is playing, and to passing whatever finger over the thumb, noticing that the thumb is placed on the key preceding the sharp or flat or else immediately after. By this means you will have as many fingers available as you have notes to play.[125]

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[121] Nicolo Pasquali, *The Art of Fingering the Harpsichord; Illustrated with Examples in Notes; to which is Added, an Approved Method of Tuning this Instrument* (Edinburgh: R. Bremner, 1760), 9.

[122] Ibid., 19.

[123] Jacquelyn DeNure McGlynn, "Keyboard Style in Late Eighteenth-Century England: A Study of Fingering, Touch and Articulation" (Master's Thesis, University of Western Ontario, 1999), 45.

[124] Ibid., 56.

[125] Johann Christian Bach and Francesco-Pasquale Ricci, *Méthode ou recueil de connaissances élémentaires pour le forte-piano ou clavecin*, (1798), trans. Athina Fytika (Genève: Minkoff Reprint, 1974), 11.

At time the treatise was written, J.C. Bach resided in London; nevertheless the treatise is in French and was composed with Ricci for the Conservatory of Naples.[126] The fingering instruction occupies only a small section of the introduction in the form of rules, but it certainly reflects the new approaches of that time.

A number of other English-language keyboard tutors also include sections on fingering, with loosely organized material and inconsistencies that reflect the transitional character of the period. In this category belong Domenico Corri's *A Complete Musical Grammar* from 1787, Augustus Frederic Christopher Kollmann's *The First Beginning on the Piano Forte* from 1785, and Nicolas-Joseph Hüllmandel's *Principles of Music, Chiefly Calculated for the Piano Forte* from 1795.[127]

There are also sources that do not include any textual explanation, but rather attempt to teach fingering through an abundance of musical examples, such as James Hook's *New Guide di Musica being a Complete Book* from 1795 and the anonymous *Music Made Easy…on the Piano Forte or Harpsichord* from 1797.[128]

Discussion

The transitional elements of the period are evident in all the sources examined, with no exceptions. The authors seem to lean toward a new approach that takes advantage of the full potential of the human hand; however, they still acknowledge the fact that the majority of the keyboardists of their time had been trained in a different system. At times this leads to suggestions that seem contradictory, but this is to be expected in an era during which a major technical change is taking place.

In spite of the increasing predominance of the fortepiano, the treatises of this period still appeal to a wide variety of keyboard instruments. For the first time in history the term fortepiano exists in some of the titles, but never exclusively. The German treatises in particular use the generic term "Klavier," which could mean any keyboard instrument. Perhaps the most common link among the sources is the constant preoccupation of the authors with providing advice not just to students but also to

[126] Ibid., 1.
[127] Jacquelyn DeNure McGlynn, 55.
[128] Ibid., 61-63.

teachers, who undoubtedly were undergoing a transition from the specific technique from their own training to the newer technique being developed at that time. This advice, while at times seemingly trivial, was imperative.

As Augustus Frederic Christopher Kollmann (1756-1829) wrote in T*he First Beginning on the Piano Forte*:

> Whether it be proper to mark the fingers over a lesson for beginners, it must be observed: that in general it is very improper to mark all the fingers, as that method doubles the objects of attention, takes away a great part of the natural ease of playing, and hinders the scholar in learning to finger with judgment.[129]

[129] Augustus Frederic Christopher Kollmann, *The First Beginning on the Piano Forte.* (London: Corri & Dussek, 1795), 11.

SOURCES FROM 1800 TO 1840

Introduction

The quest of individuals for political and spiritual independence during the Enlightenment was not realized through peaceful and rational negotiations. The last quarter of the eighteenth century was marked by two revolutions targeting political and social oppression: the American Revolution that led to the independence of the United States and the French Revolution against monarchy changed the political scenery for the upcoming century.

From the beginning of the nineteenth century the predominant philosophical idea was the view of the human as a unique individual, whose "art and religion constitute the highest manifestation of human experience."[130] Artistic expression through this search for individuality became more emotional than ever before.

According to Philip Downs,

> While the revolution made monarchs and Governments tremble, the effect upon the younger intellectuals and artists was exhilarating . . . these were the artists that called themselves romantics and saw themselves as fundamentally different from their forefathers in both ideals and actions.[131]

Artists, and particularly performing musicians, began to gain unprecedented respect and independence through their ability to express the wealth of human emotion. The popularity of public concerts in the last few decades of the eighteenth century continued and increased at the beginning of the nineteenth. Advanced technical requirements in musical compositions and the desire to attract an increasingly larger audience combined to generate a large number of virtuoso players.

Keyboard virtuosity was enhanced also through the mechanical advancements of the fortepiano, a keyboard instrument that was developing in both range and dynamic diversity. More keyboard composers wrote exclusively for the instrument, taking advantage of its full idiomatic potential, thus requiring high technical skills.

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[130] Douglass Seaton, 277.
[131] Philip G. Downs, 339-340.

The foundation of organized musical institutions like Paris' Conservatoire National de Musique in 1795 not only reflected the need for professional technical preparation of the performers, but also contributed to the systematic pedagogical presentation of required technical skills. This was the first musical institution in the modern sense that was established without charitable motives or church affiliation.[132]

Downs explains that, with its establishment, the Conservatoire

> gave many of the leading musicians of Paris a new respectability, transforming them from simple instrumentalists or composers to teachers of a noble art, with the authority of state behind them . . . the curriculum of the institution was drawn up on the theory that the art itself implies certain ideals and standards which must be propagated, and that the student had to be initiated, as it were, into the great tradition.[133]

The ideal of virtuosity together with the institutions that required systematic methodologies, led to the new generation of didactic keyboard material. Keyboard tutors from the first part of the nineteenth century differ significantly from treatises of the previous periods in their origin, organization of material, and most importantly, in their philosophy of goals.

While writers from the second half of the eighteenth century aimed to provide both students and teachers with reference tools that explained the rudiments of music, performance practice issues, and all the factors involved in technical decisions (such as ornamentation and fingering), the new era seems to have a more practical orientation. Even though some basic principles of music are still covered, a great portion of the new generation of keyboard tutors is devoted to specific exercises aimed at helping the student to acquire the utmost finger dexterity. The written text is gradually minimized in an effort to create lesson books instead of reference texts.

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[132] William Weber, "Conservatories," in *The New Grove Dictionary of Music and Musicians,* ed. Stanley Sadie (London: Macmillan Publishers Limited, 2001), Vol. 6, 314.

[133] Ibid, 637.

Muzio Clementi (1752-1832): *Introduction to the Art of Playing the Piano Forte*

Clementi's *Introduction to the Art of Playing the Piano Forte,* published in 1801, is the first book dedicated exclusively to the teaching of piano forte.[134] The format of the book is not dissimilar to tutors of the previous century, since it refers quite extensively to the rudiments of music and other theoretical aspects of playing the pianoforte.

It does differ from previous tutors in format though, since it offers a series of "pieces by various composers and of varying grades of difficulty."[135] Clementi arranged the pieces in groups of keys and composed a prelude for each set. Even though the treatise is not original in its overall format, it introduced a new approach to teaching fingering.

> To produce the BEST EFFECT, by the EASIEST MEANS, is the great basis of the art of fingering. The EFFECT, being of the highest importance, is the FIRST consulted; the WAY to accomplish it is then devised; and THAT MODE of fingering is PREFFERED which gives the BEST EFFECT, tho' not always the easiest to perform.[136]

In order for a student to achieve the ability to perform regardless of individual difficulties, Clementi focuses his fingering approach on presenting the scales, fully fingered, with patterns similar to the modern, urging their "daily practice."[137] In addition, he uses an abundance of examples to demonstrate changing fingers on repeated notes, finger substitution, broken chords and so forth. For opening the hand, Clementi introduced the arpeggiated diminished seventh chord, which "has since played an important role in many schools of technique that stress independence and equality of the fingers," as noted by Sandra Rosenblum in *Performance Practices in Classic Piano Music*.[138] The whole fingering section is dominated by exercises consisting of fragments of scale patterns and arpeggios in order to establish solid finger independence.

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[134] Roger Crager Boardman, 43.

[135] Leon Plantinga, "Introduction" in Muzio Clementi, *Gradus ad Parnassum* (New York: Da Capo Press, 1980), 3.

[136] Muzio Clementi, *Introduction to the Art of Playing on the Piano Forte* (New York, Da Capo Press, 1974), 14.

[137] Ibid., 15.

[138] Sandra P. Rosenblum, *Performance Practices in Classic Piano Music* (Bloomington: Indiana University Press, 1988), 203.

> [Clementi] provides fingering drills with scales in all major and minor keys "which ought to be practiced daily."[139] This is the first time that scales in all keys are grouped together and used as a basis for fingering. This was a new approach at the time and retains its importance even today.[140]

Exercising the hand and fingers in certain ways was the foundation of the "hand gymnastics"[141] approach to piano teaching that developed all the way through the nineteenth century. In the spirit of this approach, Clementi suggested strengthening the weaker fingers through exercises that involved holding down certain keys and repeating the same notes with the fingers that were available.

Clementi's *Introduction* was very influential since it was published in eleven editions and was translated into French, German, Spanish and Italian. He composed various pedagogical pieces, such as the *Six Sonatinas op. 36*. His most ambitious pedagogical accomplishment was *Gradus ad Parnassum*, "a compendium of one hundred piano compositions of widely differing styles issued in three volumes in 1817, 1819 and 1826."[142]

The work was not in the form of a treatise, therefore no fingering section is included; but it clearly demonstrates Clementi's pedagogical approach. The three volumes include pieces that vary in character and style, such as fugues, canons, and preludes, the vast majority of which are pianistic exercises addressing individual technical challenges, such as solid and broken octaves, double thirds, Alberti bass and so forth. The *Gradus ad Parnassum* received highest praise from critics:

> More than any of his other labours, [*Gradus ad Parnassum*] will hand his name down to the children of our grandchildren . . . will form a guide to the students of every country, in the present as well as future ages; like Bach's works it will stand as a record of the attainment in pianoforte playing, and, indeed, of the harmonic knowledge possessed by the living generation.[143]

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[139] Muzio Clementi, as quoted in Jacquelyn DeNure McGlynn, 88-89.
[140] Jacquelyn DeNure McGlynn, 88-89.
[141] Roger Crager Boardman, 43.
[142] Leon Plantinga, 4.
[143] Repository of the Arts, Series III, vol. IX (1827), 53-4. as quoted in Leon Plantinga, "Introduction" in Muzio Clementi, *Gradus ad Parnassum* (New York: Da Capo Press, 1980), 4.

Johann Nepomuk Hummel (1778-1837): *The Art of Playing the Pianoforte*

Hummel's three-part piano tutor was published in 1828. The first part explains in great detail the rudiments of music, the second part is exclusively dedicated to fingering and the third part discusses ornaments, performance practice issues, use of pedal, differences between piano makers, and even tuning.

This treatise, though following the technical ideas of Clementi, is significantly more organized in a logical manner of technical progression. The fingering section is divided into separate chapters, dealing with progressively harder issues, such as five finger position, pivoting role of the thumb, scales, finger substitution, skips, same key repetitions, crossing of the hands and so forth. For each section very detailed fingered examples demonstrate the various points. The section that Hummel uses as a vehicle to build a solid technical background is the scale section.

Not only does Hummel demonstrate all major and minor scales in one octave, all in modern fingering, he also presents scales in multiple octaves as well as in ninths, tenths and contrary motion. In addition, he suggests practicing major and minor scales in thirds and in sixths and includes the fingering for chromatic thirds.

Hummel's method includes a large number of exercises, most of them based on scale patterns, following Clementi's philosophy of training fingers through a hand gymnastics approach.[144] While Clementi merely introduced this concept, Hummel develops it to the point where the achievement of fine piano technique was considered an end in itself.[145]

Hummel very rarely provides alternative fingering in his examples. His view of the importance of the thumb is evident from the introduction of the fingering part:

> The thumb is the most important of the fingers, it is the pivot or point of support about which, whether the hand is to contract or to expand, the other fingers must turn, and direct themselves with the utmost possible facility and quickness, and without the least audible separation of the sounds.[146]

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[144] Roger Crager Boardman, 53.

[145] Adolph Kullak, *The Aesthetics of Pianoforte Playing* (New York: G. Schirmer, 1885), 16.

[146] Johann Nepomuk Hummel, *The Art of Playing the Piano Forte* (London: George Manry, 1827), 61.

However, demonstrating a conservative approach which was rare for keyboard instruction in that period, he includes a section that discusses the passing of a long finger over a shorter one, though he does not encourage the frequent use of this approach.

> Both [the passing of a long finger over a shorter and the passing of a short finger under a longer one] are to be considered as subsidiary means of accommodating the hand, by occasionally saving the too frequent passage of the thumb under the fingers; but they must not be anxiously sought after, and must always be employed in the right place.[147]

On the other hand, he was the first advocate of the general use of the thumb on the black keys.

> Before Bach, and even since his time, the thumb was scarcely ever, and the little finger but seldom used on the black keys; for which reason the compositions of that day, though easy in comparison with ours, presented great difficulties to the performer. The present style of writing renders their employment on the black keys absolutely indispensable.[148]

Carl Czerny (1791-1857): *Complete Theoretical and Practical Pianoforte School*

With Czerny's *Complete Theoretical and Practical Pianoforte School, Op. 500* of 1837, finger gymnastics, through systematic repetitive practicing of specific patterns, reached its peak.[149] The three-volume work is Czerny's most substantial theoretical accomplishment, covering an extraordinary range of topics such as improvisation, transposition, score-reading and piano maintenance. A fourth volume entitled *The Art of Playing the Ancient and Modern Forte Piano Works* was added in 1846, including advice on the performance of new works by Chopin, Liszt and others.

Fingering is clearly of utmost importance for Czerny, since he devotes an entire volume to this subject. His general recommendation is to choose fingering that is practical, simple and convenient. He is the first to "forbid finger passing"[150] (crossing

[147] Ibid., 237.
[148] Ibid., 224.
[149] Adoph Kullak, 75.
[150] Roger Crager Broadman, 65.

long over short fingers) after a long transitional period of tolerance, but is still reluctant to use the thumb and the little finger on black keys:

> When several keys are to be played one after another, either in ascending, or in descending, and that [sic] five fingers are not sufficient for this purpose, the four longer fingers must never be turned over one another; but we must either pass the thumb *under*, or pass the three middle fingers *over* the thumb.[151]

He promotes fingering for all the technical elements that were required in early nineteenth century piano playing: glissandi, chromatic runs, and note repetitions. Like Clementi and Hummel before him, the main focus of his fingering method is the scale section, where he promotes the memorization of scales and their repetitive practice on a daily basis.

According to Boardman:

> Part of the importance of scales was due to the fact that all rules for developing correct fingering, good position, a fine touch, good quality of tone, and quickness and style of execution could be developed, explained, and reduced to practice during the study of scales, so that observance of them could become a fixed habit.[152]

Czerny creates exercises that become progressively harder, beginning from a five finger position and developing into playing thirds, sixths and various scale-wise progressions. His insistence on the importance of practicing exercises is clearly evident:

> The diligent practice of finger exercises and scales, [sic] is of the highest importance; for the quick *perception* of the different values of the notes requires only a practiced eye; while for the rapid and correct *execution of them*, we also require a well-practiced finger.[153]

Czenry's teaching reflects the valuable advice of an experienced teacher with a clear understanding of the finger mechanism. He cautions against unwelcome movements of wrist and elbow while practicing arpeggios in multiple octaves, and

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[151] Carl Czerny, *Letters to Young Ladies,* (Boston: Oliver Ditson & Company, 1861), 18.
[152] Roger Crager Boardman, 66.
[153] Carl Czerny, *Letters to Young Ladies*, 15.

emphasizes control of arm weight when performing rapid arm movements, such as hand crossing:

> The crossing hand must be held very lightly, never to fall with too great weight upon the keys, so that even in quickest movements we may always retain every degree of power necessary.[154]

Czerny is undoubtedly one of the most prominent figures in the technical development of every aspiring pianist even to the present day, due to the overwhelming number of invaluable technical exercises he composed. Each study usually focuses on one particular technical aspect faithful to Czerny's ideal of achieving finger dexterity.

Other sources from this period

Clementi, Hummel and Czerny were the principal figures in the evolution of the gymnastic approach to teaching fingering. In their treatises the systematizing of the proposed exercises and methods is clearly seen. However, the increasing number of aspiring pianists, both professional and amateur, the founding of conservatories, and the expanded technical demands of pianoforte music generated a number of keyboard tutors by others.

While including some general information about music, or basic principles of piano playing, the majority of these deal to a great extent with exercises that in a scale-wise manner aim to develop finger dexterity. There is also a huge increase in published collections of etudes with the same principal concept, but in a larger and more elaborate form.

In this spirit, Johann Baptist Cramer published his *Instructions for the Piano Forte* in 1812, Friedrich Kalkbrenner published the *New Method of Studying Piano Forte* in 1837, August Eberhard Müller published an extended version of Löhlein's *Klavierschule* (naming it *Klavier und Fortepiano-Schule*) in 1804, and Friedrich Starke published the *Wiener Piano Forte-Schule* in 1819.[155] There are other similar publications

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[154] Carl Czerny, *Complete Theoretical and Practical Pianoforte School.* (London: R. Cocks and Company, 1839) Vol. I, 142.

[155] Sandra P. Rosenblum, 485-496.

from various countries, but examination of them is beyond the scope of this treatise. Louis Adam's (1758-1848) *Méthode de piano,* though, deserves special consideration because of its association with the Paris Conservatoire.

The *Méthode de piano* was published in 1804. Since Adam was a teacher at the Paris Conservatoire from 1797 until 1842, his method was probably standard pedagogical material for a large number of French-trained pianists at the beginning of the nineteenth century. Similar to other treatises of the time, Adam's work is separated into chapters dealing with rudiments of music, posture, pedaling, stylistic issues and ornamentation. The fourth chapter is concerned with fingering; it begins with an introduction, examining the principles of fingering in conjunction with observations on the construction of the human hand:

> Inspecting the hand we observe three fingers longer than the others; on the keyboard we equally observe keys more or less elevated; it's due to the conformation of the hands and the disposition of the keys that the principles of fingering are established.[156]

Certain general fingering principles are provided, but for the most part the fingering section is devoted to scale fingerings, with an abundance of exercises related to scale-like patterns. The patterns include scales in contrary motion, thirds, sixths, and other technical features found in the other treatises of the time. Adam's enhanced virtuosic aspect is underlined even more with the inclusion of practicing suggestions for extended passages with double trills.

Discussion

The beginning of the nineteenth century saw a remarkable shift in the approach to piano teaching. The independence and agility of the fingers became so important that teachers systematically provided methods in the form of exercises in order to achieve the required dexterity. The old paired fingering is almost completely abandoned and by the middle of the nineteenth century is even strongly discouraged.

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[156] Louis Adam, *Méthode de piano*, (1804), trans. Athina Fytika (Genève: Minkoff Reprint, 1974), 5.

Teachers seem to have established through their practicing and teaching that diligent scale practice ensures a solid keyboard technique. The use of all keys and the infinite number of exercises that derive from the various scale and scale-related patterns provide the basis not only for short exercises used as examples in tutors, but for even larger-scale works that are found in etude compilations.

Some of this material has been criticized occasionally for a lack of compositional imagination. Never before in keyboard history had such a large number of pieces been composed for the sole purpose of virtuosic achievement. Despite their disputed artistic value, most of these etudes have managed to maintain their value as pedagogical tools throughout two centuries of considerable advancements in piano compositions and changes in the piano itself through the development of the piano industry.

Perhaps the main appeal of the finger gymnastic approach is its clear goal: For each etude there is a particular and distinguished technical purpose, clearly identifiable by both the teacher and the student. Additionally, that purpose is achieved in a logical manner through simple and predictable harmonic progressions. The simplicity of the concept disassociates the finger motion from complex musical perceptions; therefore the technical result is direct and evident.

CHAPTER 3

ROMANTIC PERIOD

SOURCES FROM 1840 TO 1900

Introduction

The quest for individuality in all aspects of life characterized the beginning of the nineteenth century. Despite all efforts, however, the revolutions targeting political oppressions at the end of the eighteenth century did not result in a better life for most people. The industrial revolution provided luxury goods at low cost for the middle class, but also exploited the working class.

Artists and especially writers, seeking relief from their ugly and shallow surroundings, returned to Middle Age images of noble knights and gracious ladies along with ideals of unsurpassed emotionalism and self sacrifice. Willi Apel argues that, particularly for musicians, the nineteenth century was characterized by a general attitude of "longing for something nonexistent," a propensity for dream, and fancy for unrestrained subjectivism and emotionalism.[157]

During the Romantic era musicians wished to explore the whole range of emotional possibilities through their works. The strength of their expression derived from their emotional experience, which subsequently generated a chain of emotional reactions experienced by their audiences. It is no coincidence that during the nineteenth century a whole generation of virtuosos, especially pianists, allured audiences with their highly emotional interpretations.

This emotional reaction was accomplished through sensitivity in playing, as well as a great variety of special pianistic effects such as tremolos, rapidly repeated chords and consecutive octaves that increased the range of volume and timbre of piano sounds.[158] All these devices increased the already high standards of technical proficiency for

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[157] Willi Apel, *The History of Keyboard Music to 1700,* trans. and revised by Hans Tischler (Bloomington: Indiana University Press, 1972), 230.

[158] Douglass Seaton, 308.

pianists. However, the intense focus on the technical aspects of a musician's preparation, even though highly expected, was not in accordance with romantic ideals.

Douglass Seaton comments:

> Virtuosity in and of itself is "antiromantic" . . . Thus, while empty virtuosity contradicts romanticism, the cult of the virtuoso actually represents a manifestation of Romanticism in nineteenth century life.[159]

A wide range of pianistic styles was produced by this cult. Contrary to a number of virtuosos who aimed for impressiveness and showmanship, the pianists and composers whose reputation lasted through time used technical virtuosity only as a means of expression.

> The best composers and players of piano music in the nineteenth century made constant efforts to avoid the two extremes of sentimental salon music and pointless technical display.[160]

Perhaps it was the fear of generating pianists whose sole aim was technical display that caused the gradual disappearance of piano tutors, at least in the form in which they existed until the beginning of the nineteenth century. A number of etude compilations appeared, following in the steps of exercise books by Clementi, Hummel and Czerny. Despite their didactic purpose they are not comprehensive in nature, but rather aim to serve as lesson and technical books; hence they give no additional information on the philosophy behind their fingering suggestions. The fear of over-emphasizing technique in piano playing is demonstrated in Adolph Bernhard Marx's *Ein Wink für Klavierspieler*, as described by Adolph Kullak:

> In the technical virtuosity of modern times Marx finds a deficiency, the individualization of fingers not being satisfactorily developed. This is not to be understood as disallowing the independence and gymnastic training of the same; these are admitted; what the fingers lack is the inspiration of the tone. He might have expressed himself simply as follows: Modern players lack that psychic element which perceives and develops the poetic charm in the production of the single tone.[161]

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[159] Ibid., 308.
[160] Donald Jay Grout, 560.
[161] Adolph Kullak, 84.

The piano method by Ignaz Moscheles and François Joseph Fétis is an exception to this rule; therefore it will be examined below in greater detail. In addition, the only two major theoretical piano books of the nineteenth century will be discussed: Adolph Kullak's *Die Aesthetik des Klavierspiels* and Mathis M. Lussy's *Traité de l'expression musicale*.

Unlike previous periods where composers or teachers themselves felt obligated to contribute to keyboard training by compiling their accumulated pedagogical knowledge, the great teachers from the Romantic era did not write piano tutors. However, there are accounts and methods by their pupils that reflect their ideas, some of which include inconsistencies and/or conspicuous points, leading contemporary musicians to the conclusions that "methods are usually made up by the less talented students of a great teacher."[162] The teachings of Theodor Leschetizky and Ludwig Deppe, two of the most famous piano teachers of the nineteenth century, influenced many prominent performers of the Romantic era; therefore their opinions on fingering will be examined in this chapter.

Finally, the nineteenth century saw the birth of the concert etude. Frédéric Chopin was the first to give this genre its complete artistic form, "a form where musical substance and technical difficulty coincide."[163] Certain assertions on how this genre changed the perspective on fingering can be derived from the study of selected etudes by leading Romantic composers.

Comprehensive piano methods and treatises

François Joseph Fétis (1784-1871) and Ignaz Moscheles (1794-1870) together published the *Méthode des méthodes de piano* in 1840. The awareness of an overwhelming number of already existing piano methods is evident from the *Méthode*'s introduction:

[162] David Dubal, *Reflections from the Keyboard* (New York: Schirmer Books, 1977), 201.
[163] Charles Rosen, *The Romantic Generation* (Cambridge: Harvard University Press, 1995), 363.

> Many piano teachers are convinced that there is nothing more to say of playing this instrument: to them, I answer with an invitation to read my work; if they persist in their opinion after examining it with care, I will acknowledge readily that I was wrong to create it.[164]

The *Méthode* is organized in the familiar manner of initially introducing rudiments of music while practical aspects of piano playing are analyzed afterwards. It is divided into chapters with clear headings and accompanied by an abundance of examples. One of the treatise's most important contributions is its detailed references to both contemporary and earlier sources, found not just in a single prefatory reference, but in detailed sections which examine the differences among other authors and their technical approaches.

In the fingering section, the most extended part of the work, the discussion begins with some basic principles, such as consideration of the musical context in deciding on the fingering of a passage. The bulk of the fingering discussion concentrates on scale or scale-wise patterns that need to be practiced thoroughly in order for the pianist to acquire the desired technical facility.

The explanations for the fingering decisions are very thorough. In addition, there are many examples and references to the performance practices of famous pianists. For example, when discussing the performance of extended passages of consecutive octaves, Fétis and Moscheles suggest alternating between the fourth and fifth finger for the execution of the top octave note.

> The best way to execute these passages would be the fingering that M. Kalkbrenner suggested [Kalkbrenner suggested the alternation of the fourth and fifth finger], if this fingering has the power which is very often necessary for such passages . . . Mr. Liszt executes chromatic sections in octaves without this fingering with a rapidity derived from a prodigy.[165]

The discussion involves all major technical features such as arpeggios, chords (blocked and rolled), hand crossing, and references to possible exceptions of the standard rules due to specific requirements of individual compositions. A set of etudes follows

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[164] François Joseph Fétis and Ignaz Moscheles, *Méthode des méthodes de piano,* trans. Athina Fytika (Genève: Minkoff Reprint, 1973), preface.

[165] Ibid., 56.

and concludes the method. The work reflects the same spirit as Hummel's and Czerny's exercises and is thorough and useful due to its references to other methods and performance practices.

Numerous additional methods published throughout the nineteenth century use the format of concise verbal explanations followed by an abundance of exercises for developing finger independence. Some typical and widely used ones are the various didactic works by Johann Baptist Cramer, Louis Plaidy's *Technische Studien* from 1852, Louis Köhlers' *Systematische Lehrmethode für Klavierspiel und Music,* Hugo Riemann's *Vergleichende theoretisch-praktische Klavierschule,* and Hanon's *Le pianiste virtuose.*

The Aesthetics of Pianoforte Playing by Adolph Kullak (1823-1862) was first published in 1861. Later publications included supplements by the editors, Hans Bischoff and Walter Niemann.[166] This is perhaps the most important theoretical pianoforte treatise of the nineteenth century. It consists of two parts, the first a historical overview of pianoforte methods, and the second ("The beautiful in pianoforte playing") a discussion of technical matters.

The book provides a comprehensive overview of the history of pianoforte treatises, starting from C.P.E. Bach's *Versuch* and covering other treatises until the end of the nineteenth century. The analysis of each treatise includes not merely a description of its contents, but also a discussion of major pedagogical ideas and changes in terms of technical and aesthetic approaches. In this sense the work is unique because, unlike all other methods, it explores the idea of a universal knowledge.[167]

Kullak, in his discussion of technical matters, reflects a concern for the quality of tone production rather than focusing on a discussion of purely finger training. Fingering references appear throughout the discussion of technical preparation rather than in a separate discussion of fingering. Very few musical examples are provided, but there are many descriptive suggestions for practicing, similar to those seen in other nineteenth century methods, such as holding down notes and repeating others for the strengthening of fingers.

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[166] Elena Letňanová, *Piano Interpretation in the Seventeenth, Eighteenth and Nineteenth Centuries: A Study of Theory and Practice Using Original Documents* (Jefferson: McFarland & Company, Inc., 1991), 141.

[167] Ibid., 142.

Kullak's *Aesthetics* advocates the independence of all fingers, but does not consider them equal by nature. He acknowledges that the third is the strongest, while the second, fifth and fourth follow. The thumb is discussed separately due to the peculiarities of its shape and strength. This characterization of the individual fingers is not casual; Kullak demonstrates an impressive understanding of the physiology of the human hand and justifies his assertions on finger functionality.

For instance, in discussing the strength of the fifth finger, Kullak states:

> Shorter and weaker than the former [the second finger], it requires double perseverance for its strengthening; in addition, its somewhat straighter tip-joint allows from the outset less pressure on the key than is exerted by the central fingers. Consequently its lift before striking must be higher, to attain a correspondingly greater pressure.[168]

Kullak refers to standard nineteenth-century fingering approaches, such as the pivoting role of the thumb, but he is more flexible in his views than other writers from the beginning of the century. He acknowledges the possibility of passing long fingers over shorter ones, or even the possibility of passing longer fingers under shorter ones in extreme circumstances. He refers to the importance of scale practice in the pianist's routine, but recommends practicing actual compositions concurrent with mechanical practice. The enormous significance of Kullak's approach is his justification for all technical suggestions, such as the fingering of thirds, chromatic scales, glissandi, octaves and various jumps.

Elena Letňanová in *Piano Interpretation in the Seventeenth, Eighteenth and Nineteenth Centuries: A Study of Theory and Practice Using Original Documents* evaluates Kullak's book:

> Manuals of piano methods do not like to explain why certain ways or techniques are more progressive than others or more rational in their evolution. A typical characteristic is their emphasis on the external appearance of the move in apparatus or the player's arms and hands – not on the internal state of the interpreter.[169]

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[168]Adolph Kullak, 116.
[169] Elena Letňanová, 143.

Mathis Lussy's (1828-1910) *Traité de l'expression musicale* from 1873 is the major French theoretical treatise on music from the nineteenth century. It was highly respected at the time of its publication due to its insightful perspective on the appropriate aesthetic and stylistic interpretation of music. Even though it includes extended references to piano music and its interpretation, it focuses on vocalists and instrumentalists in general.

As a result, the treatise does not include any practical information on purely technical keyboard skills such as fingering. Nevertheless, the *Traité de l'expression* declares the increasingly felt need, towards the last quarter of the nineteenth century, for a disassociation of musical performance from its purely technical achievement. In his preface Lussy underlines the need for a work that discusses musical aesthetics rather than one that adds nothing inventive to the already large number of works devoted to technical exercises.

> The popularization of music has of late made astonishing progress, and yet Expression – the essence of music – seems to remain the property of a few gifted spirits, and brilliant execution is still far oftener met with than expressive playing.[170]

Ludwig Deppe and Theodor Leschetizky

Ludwig Deppe (1828-1890) was a famous pianist, teacher, conductor, and composer.[171] He taught many celebrated pianists, including Emil von Sauer. His method of teaching was considered revolutionary at the time since he was the first to advocate careful attention to muscular movements. Unfortunately, he did not write a teaching manual. His approach to piano survived through testimonies from his students, and particularly through Amy Fay's writings in *Music-Study in Germany* from 1880. Deppe was the first to develop a scientific theory of teaching piano with a unified muscular and mental coordination.

Roger Boardman summarizes Deppe's method:

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[170] M. Mathis Lussy, *Musical Expression* trans. M.E. von Glehn (London: Novello and Company Ltd., 1892), iii.

[171] John Warrack, "Deppe, Ludwig" *The New Grove Dictionary of Music and Musicians,* ed. Stanley Sadie (Washington D.C.: Grove's Dictionaries of Music, 1980), Vol. 7, 224.

> [Deppe] believed that a wholesome distribution of effort over every part of the mechanism from shoulder to finger tips was the solution to the problem of gaining mastery of the playing technic [sic]. His first step in achieving this end was to free the pianist from the well established tradition of maintaining a mechanism in which fingers moved while all else remained quiet.[172]

According to his students, Deppe recommended piano playing with the tips of the fingers, and low finger motions. In addition, he preached the coordination of the finger muscles with the muscles from the hand, arm and upper body.[173] Essentially, his teaching explained and urged the participation of the muscular system of the whole upper body for every motion realized by the fingers.

For example, on the passing of the thumb under the third finger, Fay mentions that:

> His principle in playing the scale is *not* to turn the thumb under! but to turn a little on each finger end, pressing it firmly down on the key, and screwing it round, as it were on a pivot, till the next finger is brought over its own key. In this way, he prepares for the thumb, which is left free from the hand and slightly curved.[174]

Even though there are no detailed accounts of Deppe's particular fingering suggestions, his teaching method marks a definite shift in the perception of the finger mechanism and finger usage. He did not believe in the separation of technical and musical preparation, though according to his students, Deppe did suggest some preparatory exercises, such as holding down some keys and repeating others with the remaining fingers.

However, his main concern was not the strengthening of the fingers, but rather the control of tone quality. His idea that all muscles could be controlled by the performer's mind into an accomplished synchronized action was the foundation of his perception that each finger has the potential to be equal and independent.

> Through his theory of "equal rights," he [Deppe] abolished the idea of training the fingers themselves to hit with equal power. This theory stated that each finger

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[172] Roger Boardman, 115.
[173] Elena Letňanová, 100.
[174] Amy Fay, *Music-Study in Germany, from the Home Correspondence of Amy Fay* (Philadelphia: Theodore Presser Co., 1896), 290.

> could serve equally well as a medium for transmitting the power of the whole mechanism to the key, instead of using merely its own power.[175]

Theodor Leschetizky (1830-1915) was a Polish-born pianist and one of the first to acquire an international reputation as a pedagogue. Some of the world's most famous pianists were his students, among them Annette Essipova, Ignaz Jan Paderewski, Ossip Gabrilowitsch, Ignaz Friedman and Artur Schnabel.[176] Even though Leschetizky left no written piano method, he personally endorsed the method of his student and assistant Malwine Brée, who published *The Groundwork of the Leschetizky Method* in 1902.

Leschetizky's approval is expressed in the introduction of the book:

> As you know, I am from principle no friend of theoretical Piano-Methods; but your excellent work, which I have carefully examined, is such a brilliant exposition of my personal views, that I subscribe, word for word, to everything you advance therein.[177]

From the very beginning the Leschetizky method presents concerns that challenge the ideals of finger gymnastics. The method's initial suggestions are essentially exercises that involve holding down keys with several fingers and repeating the same note (or notes) with the remaining fingers. Even though the obvious purpose of such exercises is the strengthening and independence of the fingers, the main discussion on these exercises revolves around the desirable touch and pressure in order to execute the repetitions. In addition, Leschetizky, demonstrates an unprecedented awareness of the physiological factors that contribute to finger movements:

> Be careful not to hold the inactive fingers up spasmodically, for this would take too much strength from the active ones. And do not worry if the fourth finger jerks a little when the third finger plays, or if the fifth does likewise when the fourth plays. There is an anatomical reason for this, in the presence of a common tendon; so it does no harm.[178]

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[175] Roger Boardman, 117.

[176] David Dubal, 376.

[177] Malwine Brée, *The Groundwork of the Leschetizky Method* (New York: G. Schirmer, Inc., 1902), iv.

[178] Ibid., 11.

Despite the technical orientation of the method and the abundance of preparatory exercises, Leschetizky disassociated himself from the strictly technical aspect of many of the methods devoted solely to finger exercises.

> The devotee of the piano who treats the "dry" finger exercises disdainfully does himself the greatest injury; for such exercises are the same, for the "pianistic member," the hand, as voice development for the singer's vocal organs.[179]

His disassociation from the systematic approach of finger gymnastics methods is also evident from the lack of elaborate fingering indications in the scales presented. The method provides preparatory exercises for scales, discussing in detail the wrist and arm movements associated with the passing of the thumb under the other fingers, rather than presenting dry scale fingerings. Perhaps Leschetizky's careful study of the individuality of the human hand averted his being inclined toward any standardized patterns.

According to Ethel Newcomb, another one of his assistants:

> [Leschetizky] would discuss the hand from every point of view; what this sort of hand should do, and why another kind of hand should be held differently and should be required to do otherwise.[180]

All accounts by his students indicate that Leschetizky promoted fingering that simply serves music the best. Allegedly he once said to a pupil: "Play it with your nose if necessary, but make it sound right."[181] He was more concerned with tone color and elimination of unnecessary movements than with the application or teaching of standardized fingering. Instead, he discussed in detail the preparation of fingers along with the whole arm in order for any technical device to be played appropriately.

> Leschetizky recommended preparation as a safety device for striking chords correctly. . . . When a chord was repeated in another part of the keyboard, the shape of the hand was retained while the arm swung from the first position directly to the second. If a chord changed in structure, then the hand was to take the shape needed for the coming chord while the hand was still in the air.[182]

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[179] Ibid., 28.

[180] Harold C. Schonberg, *The Great Pianists from Mozart to the Present* (New York: Simon & Schuster, 1987)*, 295.*

[181] Seymour Bernstein, *With Your Own Two Hands* (New York: G. Schirmer, Inc., 1981), 56.

[182] Roger Boardman, 89.

Concert Etudes

The etude was one of the keyboard genres that flourished in the nineteenth century. Piano etudes with a conceptually didactic purpose existed from the eighteenth century, but it was not until the 1830s that concert etudes appeared.

According to Charles Rosen:

> The etude is a Romantic idea . . . a short piece in which the musical interest is derived almost exclusively from a single technical problem. A mechanical difficulty directly produces the music, its charm and its pathos. Beauty and technique are united, but the creative stimulus is the hand, with its arrangement of muscles and tendons, its idiosyncratic shape.[183]

Frédéric Chopin's (1810-1849) two sets of *Etudes*, Opus 10 from 1833 and Opus 25 from 1837, are certainly not mechanical finger pieces, but rather works of great musical depth. The technical challenges encountered in them require certain fingering decisions in order for the patterns to be executed. In addition, there are many notes from Chopin's own teaching that demonstrate his insightful perspective on fingering. Chopin himself believed that fingering was the basis of good playing. He intended to publish a method for his teaching ideas, but "never got beyond a few penciled pages" on this project.[184]

According to one of Chopin's students, Karol Mikuli:

> Chopin gives much more freedom to the thumb on black keys. . . . The so called Black Keys *Etude,* Op. 10, No. 5, was composed principally for the task of exercising the fingers, including the thumb, on the black keys . . . he considered it [the thumb] the strongest finger of all.[185]

Chopin's approach to fingering was contrary to the principal finger equality ideal upon which keyboard teaching was developed for a whole century before him.

Rosen observes:

> Chopin insisted that each finger was fundamentally different in character and that the performer should try to exploit that difference. His use of the fourth and fifth

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[183] Charles Rosen, *The Romantic Generation,* 363.

[184] Harold C. Schonberg, 160.

[185] Claudine Lapointe, "Chopin's Fingering [sic] and their Application to Performance of his Piano Music Today," (Master's thesis, University of California, Los Angeles, 1989), 11.

> fingers for delicate chromatic effects became almost a trademark: in fact the *Etude* "In thirds," Op. 25, No. 6 depends on this technique.[186]

The perception that different fingers have fundamentally different abilities in tone production was not the only "anachronistic" fingering approach by Chopin. The *Etude* Op. 10, No. 2, is based on the concept of passing the third, fourth and fifth fingers of the right hand over and under each other while playing chromatic scales without the participation of the thumb or the second finger.[187]

Undoubtedly many of Chopin's fingering examples are in accordance with the standardized keyboard fingering of the end of the eighteenth and the beginning of the nineteenth century. Even Chopin's "unconventional" suggestions were not original in conception since they had been in use extensively from the sixteenth until the end of the eighteenth century. What was highly original, though, was his choice of fingering based not necessarily on mechanical habits or articulation needs, but rather on the desired timbral effects.

Franz Liszt (1811-1886) taught some of the most famous pianists of the late nineteenth century. His students came from all over Europe to study with him and included such prominent figures as Carl Tausig, Hans von Bülow, Eugène d'Albert, Moritz Rosenthal and many others.[188]

Liszt never considered himself a professor, but rather a musician who could provide "advice and illustration."[189] As a result, he left no pedagogical method, nor any treatise that demonstrates his teaching ideas, although he completed twelve volumes of technical studies in 1879 that were published after his death.

In *The Liszt Studies* Elyse Mach describes allegations about the final form that Liszt intended to give to this work:

> Besides the twelve volumes of Technical Studies there was, apparently, a method book that accompanied them. Composer Camille Saint-Säens indicated that Liszt wrote a "method" which was entrusted to others and mysteriously disappeared. It

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[186] Charles Rosen, *The Romantic Generation,* 368.
[187] Claudine Lapointe, 15.
[188] Elena Letňanová, 126.
[189] Harold C. Schonberg, 256.

> has even been alleged by some that the mysterious method was taken by one of Liszt's female students.[190]

Additionally, Auguste Boissier kept a very detailed diary of the lessons Liszt gave to her daughter. The diary includes Liszt's suggestions on daily technical preparations. According to this he believed that fingers should sound "rounded and totally equal."[191] However, he acknowledged their natural differences as Boissier describes in the eighteenth lesson:

> The purpose is to make them perfectly equal and independent. The fourth, the smallest, and the third are the worst and therefore need more attention; the others however must also be developed.[192]

The secondary nature of Boissier's source results at times in considerable inconsistencies. As stated above, Liszt considered the third finger as a weak finger; however, in the sixteenth lesson Liszt allegedly characterized the third finger as "too strong." In the twenty-fourth lesson Liszt describes the thumb, third and fifth fingers as the "fundamental" ones since they are the pivots of the hand.[193]

Aside from the occasional contradictions, Liszt's preoccupation with the sufficient technical preparation of all fingers through the constant use of exercise material by Czerny, Moscheles and Kalkbrenner is evident. Like all of the important piano teachers of his time, though, Liszt's main concern was a tone quality with inherent nuance, even when he discussed scale practicing.

In his technical exercises Liszt places considerable emphasis on patterns that are played by consecutive fingers and alternating hands. His deviation from the "standardized fingering"[194] is evident through his alternative fingering for drilling

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[190] Elyse Mach, "Preface" in *The Liszt Studies* (New York: Associated Music Publishers, 1973), iii.

[191] Auguste Boissier, "A Diary of Franz Liszt as a Teacher" in *The Liszt Studies* (New York: Associated Music Publishers, 1973), xvii.

[192] Ibid., xix.

[193] Ibid., xxiii.

[194] In the current treatise the term "standardized fingering" refers to the fingering of technical patterns such as scales and arpeggios, as it was systematized by authors such as Clementi, Hummel and Czerny.

exercises such as chromatic scales and consecutive thirds. Perhaps the most peculiar suggestion is the use of all five fingers in succession when playing scales.

> Liszt's playing a scale with all fingers in succession—12345, 12345—enabled him to reach extraordinary velocity, a smear like glissando: the trick consists of a rapid shift of the hand at the end of each group of five between the fifth finger and the thumb on the next note. It was the variety of touch that Liszt extended.[195]

Aside from his technical exercises, which do not have any genuine musical value, Liszt paid higher tribute to the concept of concert etudes than Chopin did. He composed twelve *Études d'une execution transcendante*, which he revised three times, increasing the technical difficulties with each revision, six *Études d'une execution transcendante d' après Paganini*, one *Grande étude de perfectionnement* and five *Études de concert*.

All of these etudes are extremely technically demanding, demonstrating a keyboard writing that is highly virtuosic and exploring capacities of the instrument that are introduced for the first time.[196] Even though the fingering suggestions in the etudes are sparse, the technical demands dictate a fingering approach that is beyond the standardized scale-passage fingering.

In addition, the fingering suggestions that do exist demonstrate the composer's wish to take advantage of the different tone colors that specific fingerings can produce. A characteristic example of this is *Mazeppa* from the *Transcendental Etudes*. Liszt suggests a consecutive 42 fingering for double notes when he wants to produce a "martellato" effect. Rosen analyzes this particular fingering choice:

> It should be clear that any attempt to play the martellato figure with four fingers 2/4 [and] 1/3 instead of only 2/4 [and] 2/4 (as pianists often do to avoid strain on wrist and arm), is an inexcusable betrayal of Liszt's intentions . . . all these novelties seem to me to derive from his reconception of the means of execution which creates an unprecedented dramatic force.[197]

Other Romantic keyboard composers such as Robert Schumann and Johannes Brahms also contributed through their compositions and teachings to a new technical

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[195] Charles Rosen, *The Romantic Generation,* 508.
[196] Frank Eugene Kirby, *Music for Piano: A Short History* (Portland: Amadeus Press, 1995), 209.
[197] Charles Rosen, *The Romantic Generation,* 498.

approach to piano playing. Schumann was one of the first recognized composers to create substantial musical works aimed at the development of young pianists, a trend that had been neglected since the Baroque era. Schumann and Brahms both wrote etudes, even though less transcendentally difficult than Chopin's and particularly Liszt's.

There is no detailed description of Brahms' or Schumann's particular fingering technique. What is evident, though, through their works, as well as the works of the majority of Romantic composers, is the need for pianists to use their technique in a manner that would help them accomplish sensitive and colorful piano playing.

Florence May describes her lessons with Brahms:

> He did not believe in the utility for me of the daily practice of the ordinary five-finger exercises, preferring to form exercises from any piece or study upon which I might be engaged. He had a great habit of turning a difficult passage around and making me practice it, not as written, but with other accents and in various figures.[198]

Discussion

The first half of the nineteenth century highlighted the "finger gymnastic" approach as part of the technical preparation of pianists. The teaching of standardized scale or scale-like patterns was perfected through a thorough and systematic organization of particular schemes in all the possible keys and hand positions. Piano methods based on the idea of finger equality and finger independence thrived.

All of the Romantic piano composers and famous pianists were trained with the system of hand gymnastics, and they took advantage of the unprecedented virtuosity that such meticulous finger training provided. They rejected, however, the one-dimensional aspect that a gymnastic approach threatened to project in their playing. The Romantic ideal of extravagant expression of emotion could be achieved in piano playing only through an exploration of nuances and tone colors.

As a result, major piano teachers of the second half of the nineteenth century approached the teaching of fingering in a new way. They encouraged technical training with the use of already established technical exercises, but placed more emphasis on tone

[198] Michael Musgrave, *A Brahms Reader* (New Haven: Yale University Press, 2000), 133.

production than on finger dexterity alone. They began to develop an awareness of the importance of the whole muscular system of the hand and arm as contributors in the movement of fingers.

In addition, they rejected at times the standardized rules of fingering, favoring instead the creation of special effects. Instead of working against human nature by trying to make all fingers equal, they took advantage of the different tone colors that individual fingers can produce due to their inherent peculiarities in shape and strength. It was in the second half of the nineteenth century that some of the most legendary pianists acquired fame due not just to their virtuosity, but mainly to their singing tone.

The disdain that true romantics like Schumann had for the followers of a dry technical idea is best described by Joseph Weingarten:

> [Schumann] described as "insipid virtuosity" the antics of the popular nineteenth-century pianist-composers, such as Henri Herz, Franz Hünten, Karl Czerny, Friedrich Kalkbrenner, and a host of others, and wrote of them: "Before Herz and Czerny I doff my hat – to ask that they trouble me no more."[199]

[199] Joseph Weingarten, "Interpreting Schumann's Piano Music" in *Robert Schumann: The Man and His Music* ed. Alan Walker (London: Barrie & Jenkins Ltd., 1972), 97.

CHAPTER 4

TWENTIETH CENTURY

SOURCES FROM 1900 TO THE PRESENT

Introduction

The Western world entered the twentieth century optimistically due to the temporary break from wars and revolutions. Through the magnificent scientific achievements of the time, the comforts of everyday life reached an unprecedented level. The progress in medicine, the use of electricity and the revolutionary advancements in all forms of communication generated great expectations and bright hopes for the future.

Unfortunately, soon after the beginning of the new century World War I put an end to the optimism, replacing it with a sense of frustration and pessimism. Even though the basis of political and social turmoil related to the war had its roots in the nineteenth century, it wasn't until the first two decades of the twentieth century that the sense of political and social stability was demolished.

In the course of these radically changing social structures, the artistic aesthetic depended on "the recognition that consciousness itself was grounded in tension and frustration . . . artists began to question the assumption that art should purvey beauty and pleasure."[200]

Robert P. Morgan, in *Twentieth-Century Music: A History of Musical Style in Modern Europe and America,* explains that:

> The period between 1900 and 1914 is one of the most turbulent in the entire history of the arts, one that produced a series of revolutionary developments fundamentally affecting all subsequent endeavors . . . the tendency to distort objective reality in favor of a more personalized and emotionally charged vision was evident throughout the art world.[201]

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[200] Douglass Seaton, 350.

[201] Robert P. Morgan, *Twentieth-Century Music: A History of Musical Style in Modern Europe and America* (New York: W.W. Norton and Company, 1991), 14-15.

Musicians, like all other artists, conveyed their frustration by searching for new means of personalized expression through experimentation with stylistic and performance practices. As a result, from the twentieth century until the present time technical and formalistic developments were influenced by movements such as impressionism, expressionism, serialism, neoclassicism, chance music and electronic music. Musical expression had experienced diversity in the past; however, never before had Western music culture been so varied.

Paul Griffiths in *Modern Music: A Concise History* writes:

> The difference in twentieth-century music is that so many options have remained open that there is no single stream of development, no common language such as usually existed in earlier times, but an ever-spreading delta of aims and means.[202]

The piano world was certainly not unaffected by this experimentation, and newly discovered piano sonorities seemed to be the primary goal for composers. The means to the path of discovery varied considerably. Impressionists such as Claude Debussy and Maurice Ravel explored timbral and figural nuances and experimented mainly with pedaling and register, while Béla Bartók and Igor Stravinsky approached the piano with a primitivism that treated it more like a percussive instrument.[203]

Additionally, the technical requirements for performers included a variety of unconventional means. For example, Henry Cowell required pianists to produce overtones by stopping the strings with their hands; John Cage required a particular preparation of the piano by placing various materials in the strings in order to modify the sounds; and Olivier Messiaen required the electronic manipulation or amplification of piano sonorities in some of his works. In extreme cases, such as George Crumb's piano works, pianists were even required to "sing, moan, and whistle."[204]

A large amount of piano music from the twentieth century is traditional in its appearance and in what it requires of the performers. Rachmaninoff and Scriabin signified the extended height of romanticism with works written in the twentieth century; Prokofiev and Shostakovitch wrote piano music that was highly dissonant at times but

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[202] Paul Griffiths, *Modern Music: A Concise History* (New York: Thames and Hudson, 1978), 22.

[203] Glenn Watkins, *Soundings: Music in the Twentieth Century* (New York: Schirmer Books, 1995), 622.

[204] F.E. Kirby, 387.

certainly within the technical boundaries of a classically trained pianist; even composers later in the century—such as William Bolcom and Ned Rorem—explored the potential of the piano without resorting to avant-garde techniques.[205]

Within the framework of these diverse technical and expressive means, piano keyboard instruction has been equally multidimensional. The great pianists and teachers from the beginning of the twentieth century came from the great romantic piano tradition; as a result their teachings combine the quest for virtuosity with the more essential quest for beauty in tone production. A detailed discussion of the ideas of all major late romantic pianists is beyond the scope of this treatise; however, the teaching of such important keyboard figures as Josef Lhevinne, Alfred Cortot and Isidor Philipp will be examined due to their vast influence on many major twentieth-century pianists.

Expanding scientific knowledge of the human muscular system created a new generation of keyboard instruction methods. Advancements in technical demands for pianists were not the only reason for this biomechanical approach, as the invention of the piano's cast-iron frame at the end of the nineteenth century resulted in a considerably heavier action for the instrument, and it soon became apparent that finger motion alone was not sufficient to meet the needs of the increasingly challenging keyboard repertoire.[206]

As a result, a generation of pedagogues based their teachings on an advanced knowledge of the muscular system, with the goal of making piano technique more efficient and effortless. An examination of such approaches here will include the methods of Tobias Matthay, Rudolph M. Breithaupt, Thomas Fielden, Otto Ortmann, and Abby Whiteside.

Certain pieces that call for innovative approaches to fingering due to unconventional compositional methods will be discussed as examples of avant-garde trends. Finally, an examination of contemporary methods and teaching of piano fingering will include an overview of recent research regarding effective teaching of standard or alternative keyboard fingering.

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[205] Glenn Watkins, 624.

[206] Brenda G. Wristen, "Overuse Injuries and Piano Technique: A Biomechanical Approach" (Ph.D. diss., Texas Tech University, 1998), 25.

Romantic and Post-Romantic piano teaching

Isidore Philipp (1863-1958) was a famous French pianist, teacher and music editor. For years he held a teaching position at the Paris Conservatory and also taught at the American Conservatory at Fontainebleau. Among his most famous students were Aaron Copland and Guiomar Novaes.

Philipp's principal teaching ideas are reflected in his editions and particularly his pedagogical work aimed towards creating a solid technique. His *Complete School of Technic for the Pianoforte* from 1908 is essentially a volume of finger exercises with a few explanatory comments, accompanied by an abundance of fully fingered practice patterns.

To increase the flexibility and independence of the fingers, Philipp suggests a variety of exercises requiring the holding down of certain keys and repeating one or more keys with the remaining fingers. This approach is certainly not an original concept, but Philipp expands the method by suggesting the execution of melodic patterns with the fingers that are not holding keys down, instead of simply repeating the same key.

In general, his method requires finger gymnastics, with an abundance of five-finger-position exercises, scales, and scale-wise patterns. Philipp's primary original contribution is his detailed scientific approach to preparatory exercises that precede the scale section. He considers the development of the flexibility of the thumb in passing under all the other fingers (including the fifth) to be of primary importance in a pianist's technique. Therefore, his preparatory exercises include scales with paired fingering involving the thumb, in order to enhance its pivoting role. For example, each right hand scale is required to be executed with the following pairs of fingers: 12 12 12, 13 13 13, 14 14 14 and 15 15 15. The movement of the thumb is thoroughly analyzed:

> When passing [the thumb] under, the movement comes equally from the ball of the thumb and its joints. The ball moves well toward the palm as the thumb goes under, and is kept loose and flexible. As the second finger is played, the thumb moves instantly under, its tip covering the next note it is to strike.[207]

[207] Isidor Philipp, *Complete School of Technic* (Philadelphia: Theodore Presser Co., 1908), 22.

Philipp's preoccupation with the ability of the thumb to pass under any finger in any given position is also evident in his recommendation for practicing all scales using the fingering of the C major scale. Following the preparatory exercises and scale suggestions, the method continues in a traditional finger exercise manner, suggesting the practicing of scales with the standard fingerings, as well as the practicing of arpeggios, double notes, glissandi and so forth.

Joseph Lhevinne (1874-1944), a Russian-born pianist with an international reputation, was also a very well-known teacher and one of the first piano teachers at the Juilliard School of Music. He outlined his fundamentals of piano teaching in a short book entitled *Basic Principles in Pianoforte Playing,* published in 1924.

In this book Lhevinne aims to summarize the technical and musical ideals that would enable a young pianist to achieve a sufficient technique as well as a thorough understanding of the musical composition:

> Before the student even considers the matters of technic and touch, a good grounding in real musicianship is necessary. . . . I have repeatedly had students come for instruction . . . who barely knew what key they were playing in.[208]

As a result, the book recommends a thorough knowledge of harmony and a high level of ear training. Most treatises until the beginning of the nineteenth century included extended sections on the rudiments of music, since the keyboardist's training required thoroughbass study; but apparently during the course of the nineteenth century this theoretical awareness was neglected in keyboard training, and Lhevinne felt the need to underscore its importance.

Lhevinne's major teachings revolve around tone production and the ideal singing tone. There is no particular section on fingering, but there are comments and advice related to fingering and finger functions throughout the book. He considers a thorough knowledge of scales and standardized scale fingering an irreplaceable prerequisite for every young pianist:

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[208] Josef Lhevinne, *Basic Principles in Pianoforte Playing* (1924), (New York: Dover Publications, 1972), 9.

> Most pupils look upon scales as a kind of musical gymnasium for developing the muscles. They do that, of course, and there are few technical exercises that are as good, but their great practical value is for training of the hand in fingering so that the best fingering in any key becomes automatic.[209]

He suggests scale practicing that does not necessarily begin from the tonic of each scale, so that the student is able to play any given fragment of the scale in a composition. Finally, he considers scales to be the ideal vehicle for the improvement of sight reading and harmonic knowledge. As Lhevinne states, "you may have too little scale practice, but you can never have too much."[210]

Lhevinne's most scientific suggestion is his detailed description of finger motions for the achievement of the best possible singing tone. He suggests no movement of the finger above the metacarpal joint.[211] In addition, he specifies the parts of the finger that ideally should be involved in tone production:

> It is almost an axiom to say that the smaller the surface of the first joint of the finger touching the key, the harder and blunter the tone; the larger the surface, the more ringing and singing the tone. Naturally if you find a passage requiring a very brilliant, brittle tone you employ a small striking surface, using only the tips of the fingers. [212]

Alfred Cortot (1877-1962), one of the most famous French pianists of the first half of the twentieth century, was an acclaimed teacher and editor, particularly of Chopin's works. Among his students were Magda Tagliaferro, Vlado Perlemuter and Dinu Lipatti.[213]

Cortot published his *Principes rationnels de la technique pianistique* in 1928. The format of the book is not unlike many other finger-exercise books that have already been examined. However, Cortot's approach differs from them not only in the wide variety of preparatory exercises with multiple alternative fingerings, but also in the intention to arrange his material in a pedagogical manner that assures the achievement of a fundamental technique within a specified period of time.

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[209] Ibid., 11.
[210] Ibid., 11.
[211] Ibid., 12.
[212] Ibid., 18-19.
[213] David Dubal, 370.

> A preparatory period of six months is necessary for a thorough preliminary study of this collection, consisting of three quarters of an hour's work each day, and of about a month, or more accurately, thirty-six consecutive days for the preparation of each chapter; a quarter of an hour's work would be devoted regularly, apart from any other category of exercise, every day, to the preparatory chapter entitled "Daily Keyboard Gymnastics"[214]

Even though the method starts with the familiar routine of strengthening the fingers by holding some of them and repeating or playing short patterns with the others, Cortot proceeds in an unconventional (for the early twentieth century) approach for developing finger flexibility. He suggests the practice of diatonic and chromatic scales in paired fingerings, using all combinations of fingers. For example, the right hand chromatic scale is suggested to be practiced with the following combinations: 23 23 23, 32 32 32, 34 34 34 , 43 43 43, 45 45 45, and 54 54 54.

Similarly to Isidor Philipp, Cortot considers the passing of the thumb to be the cornerstone of finger technique. He mentions performance practices of past centuries, when the thumb was hardly used, but emphasizes its importance in piano playing ever since a prominent pivotal role was attributed to it.

His suggestions for enhancing the thumb's pivotal ability are at times extreme. For instance, Cortot suggests an exercise where the second, third and fourth fingers hold down consecutive keys, while the thumb oscillates at an interval of a fifth, thus requiring the wrist to perform a 90º angle movement. The reason for these exercises is for the fingers and the hand to learn to perform even seemingly abrupt movements without causing any disruption in the tone production.

> The action of the thumb in scales and arpeggios, as an agent for the multiplication of the fingers, should neither cause any inequality of tone, any modification in the position of the other fingers, nor any diminution of speed in rapid playing.[215]

Cortot's approach to scales is that of a teacher aiming to prepare students for all possible fingerings that can be used in various musical contexts. As a result, for every

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[214] Alfred Cortot, *Rational Principles of Pianoforte Technique,* trans. R. Le Roy - Métaxas (Paris: Éditions Salabert, 1928), 2.

[215] Ibid., 25.

scale in any form—whether diatonic or chromatic—and even for double notes of any given interval, he provides all possible combinations of fingerings.

> We give not only the usual fingering or fingerings of the scales, but also the variations which may be employed according to the exigencies of musical execution. . . . They all deserve to be practiced with the greatest care, as their application to the need of interpretation constantly imposes itself. It is not therefore a "school of scales" which we intend to lay down here, but a study of all the fingerings required for their execution.[216]

The number of piano methods written by famous pianists and teachers from the first half of the twentieth century is very large. All seemed to have the same goal: providing solutions for a solid finger technique in order to enable young pianists to respond to the demands of an increasingly technically challenging piano repertoire. Their methods of approaching this goal differ, since some considered the standardized patterns essential for the building of a good technique and others emphasized more an analysis of the muscular movements that enable the production of a beautiful tone. Whether using the standardized fingering or not, all of them encourage practicing a series of technical, and very often scale-like, exercises.

In addition to the three methods that were examined in detail above, special attention should be given to the technical methods by Ernst von Dohnányi and Thomas B. Knott.

Ernst von Dohnányi (1877-1960), Hungarian composer, conductor, pianist and teacher, published his most famous piano pedagogical work entitled *Essential Finger Exercises for Obtaining a Sure Piano Technique* in 1929. The book consists entirely of exercises that aim to strengthen the fingers in a manner closely related to the ideals of finger gymnastics. Dohnányi's exercises include an abundance of scale and scale-wise patterns, exercises with holding down certain keys by one or more fingers and playing various figurations with the remaining fingers, and exercises in thirds, chords and octaves.

Despite the strictly technical orientation of his method, Dohnányi in the preface of his book dismisses the idea of the persistent practice of etudes. He believes that

[216] Ibid, 40.

mechanical etudes essentially reduce the practice time of important repertoire, which in itself provides a variety of opportunities for technical practice. Therefore, his method contains only finger exercises that, according to Dohnányi, should not be practiced mechanically, but with full mental attention:

> Finger-exercises are preferable to studies ("Etudes"), if only for the reason that they can be practiced from memory, and consequently the whole attention can be concentrated on the proper execution, which is important.[217]

Thomas Knott, a prominent teacher at the Royal Academy of Music, wrote the instruction book *Pianoforte Fingering: Its Principles and Applications* in 1928. Predominantly in narrative form, this book aims to explain and provide fingering suggestions for all common technical features in piano playing.

Knott approaches fingering in a traditional and conservative manner. He suggests the standard fingerings for scales and arpeggios, explaining the fingering choices based on the sequence of black and white keys when performing scale or scale-wise patterns. Despite a lack of imaginative solutions and methods of practicing, Knott acknowledges the physiological differences among human hands, and urges pianists to choose fingering predominantly based on the specific musical requirements. In his conclusion he even abolishes the idea of fixed fingering solutions, especially for more advanced pianists:

> It is not desirable to make a fetish of "correct" fingering . . . attention to this detail should be given at the earliest stage only, when physical means are being devised and judged. As habitude of muscular sensation becomes enhanced, so must the consciousness of it recede.[218]

Bio-mechanic methods of piano teaching

Tobias Matthay (1858 -1945) was one of the first teachers to discuss in detail the use of weight in the production of tone and the importance of exertion and relaxation of the playing apparatus. He published *The Act of Touch* in 1903; then in 1932 he published *The Visible and Invisible in Piano Technique*, presenting essentially the same ideas as

[217] Ernö Dohnányi, *Essential Finger Exercises for Obtaining a Sure Piano Technique* (New York: Edward B. Marks Music Corporation, 1929), 2-3.

[218] Thomas B. Knott, *Pianoforte Fingering* (London: Oxford University Press, 1928), 22.

before but correcting elements that he believed had led to misperceptions of his earlier book.[219] In his books Matthay stresses the importance of a pianist's understanding of the precise mechanism of the piano, in order to avoid unnecessary muscle tension.

Roger Boardman, explaining Matthay's philosophy of sound production, concludes:

> When a key is depressed approximately three-eighths of an inch, the hammer is set into motion and strikes the string, producing sound. Since physical activity following this moment of sound production could have no possible effect on the sound itself, it is important for the player to stop pressing on the key at the exact moment sound is heard.[220]

Matthay believes that every pianist is required to analyze the muscles connected with each motion required for piano playing in terms of their ability to stress and relax. He rejects the method of practical teaching by simply selecting exercises and studies so that the student can obtain certain flexibility and velocity. On the contrary, he believes that this type of technical preparation may be the source of ugly "key hitting" sounds:

> No doubt it was the influence of a certain German CONSERVATORIUM that gave it [teaching the striking of keys] such wide currency . . . The deplorably evil effects of deliberately teaching key hitting have proved incredibly far-reaching and disastrous to the progress of our art. The mechanically wrong principle it involves, not only leads with absolute certainty toward paucity of tone, and evil-sounding tone; but it also renders all subtlety, accuracy and certainty of EXPRESSION a physical impossibility.[221]

In this respect, Matthay asserts that the muscular system of the pianist's hand does not have any impact on tone production until the finger touches the key. From this point the fingertip should feel the resistance of the key in order to impart the correct motion to it.[222]

The strong forearm muscles that are used to help the finger depress a key should relax and cease working at the moment that a key is depressed. The only exception to

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[219] Brenda G. Wristen, 29.

[220] Roger Boardman, 120.

[221] Tobias Matthay, *The Act of Touch in All its Diversity* (London: Bosworth and Company Ltd., 1903), 96-97.

[222] Tobias Matthay, *The Visible and Invisible in Pianoforte Technique* (London: Oxford University Press, 1932), 13.

this would be that of a note that needs to be held down, in which case the weaker intrinsic hand muscles should continue working.[223] According to the desired tone quality, the specific finger attitude is to be decided by the performer. The two general choices are the "Hammer-touch" and the "Clinging-touch":

> In the first, or the "Hammer-Touch" variety . . . a greatly curved or bent position (like the hammer of an old fashioned percussion gun) is assumed by the finger when it is raised as preliminary to the act of tone-production. . . . In the second, or "clinging" variety of touch . . . a far less curved position is assumed by the finger as a preliminary, and it may indeed be almost unbent or "flat."[224]

Matthay does not directly analyze fingering choices, but instead gives specific instructions for the use of fingers, as well as for the muscular movements of the forearm and the whole upper arm. He considers the overall issue of hand position to be overrated; instead he believes in the necessity of preparing each position. In this way the pianist can feel the key before depressing it so that the muscular exertion can be adjusted.

Timely preparation of the hand position requires preparation of the wrist and arm motions as well. Matthay believes in the total control of the exertion and relaxation of the whole muscular system. If the correct muscular conditions are set, "the details of positions would fulfill themselves."[225]

Rudolph Maria Breithaupt (1873-1945) published *Die Natürliche Klaviertechnik* in 1904. Breithaupt's method, similar to Matthay's, analyzes the movements of the fingers as a combination of muscular contractions and extensions while the hand is twisting, turning, and gliding.[226] In addition, all hand and finger motions are closely attached to the specific technical requirements of each piano that is used for practicing or performance.

Breithaupt believes that a fluent technique is based on the coordination of the muscular functions between the various parts of the hand and arm. For example, rapid extension of the forearm is necessary for shifting the thumb under the other fingers. Similarly, the distribution of weight from left to right during arm rotation is essential for

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[223] Ibid., 31.
[224] Tobias Matthay, *The Act of Touch in All its Diversity*, 109-110.
[225] Roger Boardman, 131.
[226] Rudolph M. Breithaupt, *Natural Piano Technic,* trans. John Bernhoff (Leipzig: C.F. Kahnt Nachfolger, 1909), 10.

executing technical devices such as tremolos, or broken octaves.[227] According to Breithaupt, rotary motion is also the basis for scale passages. He does not believe in the necessity of special preparatory exercises for the pivoting preparation of the thumb, since this is an action that can be achieved through rotary movement.

Breithaupt's procedure of teaching a scale is summarized by Boardman:

> Breithaupt had the student play the scale [B major] in two rotary swings, going from B to E and E to B; then one rotary swing, going from B to B, then two swings, B to B and B to B an octave higher. Finally, the two octave scale from B to B was played with one swing.[228]

The teaching of all technical patterns, such as scales, arpeggios, double notes, etc., is done through rhythmical exercises that emphasize the section of the pattern that requires coordination of muscles from the forearm and upper arm. For example, all skips need to be executed by a free swing and descent of the arm, in one curved primary movement.

Breithaupt's fingering principles fundamentally opposed the ideals of the finger gymnastic school. He believes that the most adequate fingering results from the most natural movements of the hand and arm muscles; therefore he objects to predetermined fingering systems, as well as to methods that aim for finger strengthening through high finger motions.

Instead, he teaches fingering determined by the perception of the weight and the sense of key touch. Similarly to Deppe's teaching, Breithaupt believes that the only way the fingers can acquire equal strength is to disassociate the finger motions from the individual motions of the finger muscles.[229] In this respect, the exercise of the fingers should follow a thorough study of the proper usage of arm weight.

In order for such control to be achieved, his teaching method rejects all traditional scale and finger gymnastics types of exercises. Instead, Breithaupt suggests a series of daily exercises, the aim of which is to control the muscle weight and movements. Particularly for the fingers, the exercises require the lifting of each finger one inch above

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[227] Roger Boardman, 136.
[228] Ibid, 138.
[229] Ibid., 145.

the key and the consequent falling of the finger to the key with all the arm weight transferred to it.[230]

Thomas Fielden published the *Science of Pianoforte Technique* in 1924. He is among the first piano pedagogues who explored the interrelationship between the mental, nervous and muscular factors that all contribute to piano playing. He wrote:

> The mind must have knowledge of the muscular movements which take place in any given action of the arms and fingers. . . . Nerves are the means of communication from the brain to the muscles, and need training and refining just as the muscles themselves do. . . . The muscular side is responsible for the final execution of the original mental conception.[231]

Fielden's detailed analysis includes numerous illustrations of all parts of the playing apparatus: bones, tendons, ligaments and muscles. In discussing the action of the fingers he considers that the most essential movement of the finger muscles is the lifting of the finger before depressing the key. The passing of the thumb under other fingers, though, is part of a larger action that involves the lateral movement of the forearm.

Even though Fielden believes in the coordination of the whole arm and hand mechanism, he advocates the practice of gymnastics away from the keyboard as part of one's fundamental technical training. Fielden's use of "gymnastics'" is not related to finger gymnastic exercises that had been introduced by most piano pedagogues of the nineteenth century. Instead of prescribing exercises for finger coordination, he provides separate exercises for the forearm, arm and fingers that develop an awareness of mental control of muscle movements.

Based on the principle that the greater the ease and resilience of the pianist's physical movement, the more spontaneous and expressive the performance will be,[232] Fielden believes that fingering should evolve along with the progress of muscular awareness. However, he believes that at that time, fingering had not yet evolved with the new technical ideas based on muscle coordination and relaxation principles.[233]

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[230] Ibid., 150.
[231] Ibid., 156.
[232] Thomas Fielden, *The Science of Pianoforte Technique* (London: Macmillan and Company, 1934), 168.
[233] Ibid., 168-180.

Otto Ortmann (1889-1979) published *The Physical Basis of Piano Touch and Tone* in 1925. This book is based on scientific research on the various aspects of sound production in playing the piano. Ortmann discusses in detail the vibration of the string and the production of tone. In addition, he provides graphic representations of sound waves, based on different stroke types, and describes the anatomical parts of the hand and arm and their effects on producing sound.

The book is not a practical manual of piano playing, and it lacks specific information on fingering. However, Ortmann does examine various finger positions with their possible effects on key speed, and he provides diagrams showing the direct impact of these positions on the tone produced:

> The curved finger strikes the key with its nail joint vertical. The straight or flat finger has its nail joint almost horizontal. . . . The greatest difference is found in the percussive and non-percussive elements [in the diagram of the sound wave].[234]

Ortmann continued his research and experiments on piano sound production and published *The Physiological Mechanics of Piano Technique* in 1929. Based on the principles of physics and human anatomy, it discusses the skeletal, muscular, neural and circulatory components of piano playing.[235] Ortmann demonstrates that any finger movement is a result of a combination of muscular and neurological movements of the playing apparatus; therefore, an absence of finger motion does not guarantee absence of muscular activity.

In his discussion of fingering, Ortmann suggests choices that ensure ease and smoothness of the requisite movement. He believes in the multi-dimensional character of the subject, since fingering choices should also take into account musical considerations, desired tone quality and anatomical individuality.

> The fingering of a passage should not, in many instances, be applied fixedly to all hand-types. . . . A particular abnormality, let us say, high webs between the third and the fourth, and between the fourth and the fifth fingers, will make abduction

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[234] Otto Ortmann, *The Physical Basis of Piano Touch and Tone* (London: Kegan Paul, Trench, Trubner and Co., Ltd., 1925), 23.

[235] Brenda G. Wristen, 43.

> among these fingers quite difficult, and the player frequently substitutes the third for the fourth finger in chordal structures demanding wide abduction.[236]

Abby Whiteside (1881-1956) published *Indispensables of Piano Playing* in 1955. Whiteside discusses extensively the participation of the upper arm or even the torso as providers of physical support for the fingers. She believes that the upper arm is more responsible for the key action than are the finger muscles. According to her ideas, a pianist with an awareness of the larger muscles can achieve speed and power without overburdening the smaller muscles.[237]

As a result, Whiteside does not elaborate on finger motions, since she believes that they are controlled by the larger arm motions. She is extremely critical of finger technique schools; according to her, "training the fingers for hitting strength is the basis for all 'pianists' cramp.'"[238] Her firm belief is that the fingers themselves cannot be made equally strong and they are insufficient for producing loud dynamics; therefore technical training should involve mainly the larger muscles.

Whiteside rejects the use of scales and scale-like patterns for practicing, because after a certain period they can be played easily using the fingers only; she feels this does not contribute to the ultimate technical goal: the balanced activity of the whole body. Particularly in regard to standard finger exercises such as Hanon and Czerny, she states that they should be "completely discarded" for lack of sufficient musical stimulation.[239] The book does not contribute suggestions in the area of practical fingering, since Whiteside expresses an altogether opposite position to fingering:

> I should say that the importance of a prescribed fingering is practically nil. If you avoid fussing about fingering you will never produce a lasting obstacle to fluent passage work. If a rhythm is working, a finger will be ready to deliver power.[240]

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[236] Otto Ortmann, *Physiological Mechanics of Piano Technique* (1929) (New York: E.P. Dutton and Co., Inc., 1962), 280.

[237] Brenda G. Wristen, 42.

[238] Abby Whiteside, *Indispensables of Piano Playing* (New York: Charles Scribner's Sons, 1955), 49.

[239] Ibid., 50.

[240] Ibid., 50.

There are many other detailed and experimental approaches to the biomechanics of playing the piano. Arnold Schultz's *Riddle of the Pianist's Finger and its Relationship to a Touch-Scheme* from 1932 and George Kochevitsky's *The Art of Piano Playing* from 1967 represent only two of the researchers and pedagogues who attempted a scientific approach to the matter. An analytical investigation of all contemporary views on coordination of the playing apparatus is beyond the scope of the present treatise. The selected sources are summarized in order to describe briefly the principles and results of twentieth-century research on piano technique.

Unconventional fingering instructions

One of the compositional trends in twentieth-century piano works is experimentation with "extended techniques." In an effort to extract new sounds from the instrument, composers resort to unconventional ways of playing the piano. In the course of representing their ideas they either use conventional keyboard notation with some additional instructions, or invent unique notational systems usually consisting of written directions and graphic diagrams.[241]

These alternative notational systems result in varying amounts of freedom for the performer. In cases such as Alvin Lucier's *Action Music for Piano* (1962), the pianist is asked to create sounds based on an abstract image. On the other hand, the strictness and precision in the notation of Karlheinz Stockhausen's *Klavierstück I* leaves the performer with very few interpretational choices.[242]

Nevertheless, piano works that use extended techniques are customarily very explicit in their notation and often contain detailed explanatory introductions and comments. Twentieth-century composers, perhaps rejecting the freedom that late Romantic virtuosity brought to piano performance practices, have been particularly concerned with the precision and attention to detail in the performance of their works.

Milton Babbitt expresses this concern:

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[241] F.E. Kirby, 382.

[242] Margaret Ellen Rose, "Coming to Terms with the Twentieth Century Using a Nineteenth Century Instrument: Virtuosity, Gesture and Visual Rhetoric in Contemporary Piano Compositions and Performance" (Ph.D. diss., University of California, San Diego, 1987), 62-63.

> An incorrectly performed or perceived dynamic value results [not only] in destruction of the work's dynamic pattern, but also in false identification of other components of the event . . . creating incorrect pitch, registral, timbral and durational associations.[243]

Music that incorporates many unconventional elements increases the range of the pianist's movements. Whereas traditionally the pianist concentrates on the movements of arms and hands in order to achieve the desired tone, with new techniques he or she is often required to prepare large movements involving the whole body. As a result, fingering choices are limited or determined by overall body position and are not necessarily associated with the desired tone production.

Margaret Ellen Rose writes:

> One could say that in music of this sort, with its overwhelming emphasis on timbre, the pianist "goes for the 'spot' (on the instrument)," whereas in piano music which uses only the keyboard, the pianist "goes for the sound."[244]

Henry Cowell (1897-1965) was one of the early pioneers of extended piano techniques. His *Aeolian Harp* from 1923 is essentially an experiment with the sonorities produced when a chord is pressed down silently with one hand while the other plays on open strings. Cowell does not provide specific fingering indications. However, he indicates the part of the finger that should touch the string in order to produce the desired timbre, thus requiring on-the-string action at times by either the "back of thumb nail" or "the flesh of finger."[245]

Perhaps the most common avant-garde piano technique is the use of clusters. Clusters are performed by bunched fingers, or a flat hand on the keys, or with the use of a mechanical contrivance as, for example, in Charles Ives' *Concord Sonata*, in which a wooden board is used.[246]

George Crumb (b. 1929) in his *Makrokosmos* indicates fingering whenever he requires a particular effect. For example, he suggests particular fingerings while playing

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[243] Milton Babbitt, "Who Cares if You Listen?" *High Fidelity* 8, no.2 (February 1958): 38-40, 126-27.

[244] Margaret Ellen Rose, 69.

[245] Robert P. Morgan, ed., *Anthology of Twentieth-Century Music* (New York: W.W. Norton and Company, 1992), 264.

[246] Gardner Read, *Compendium of Modern Instrumental Techniques* (Westport: Greenwood Press, 1993), 206.

on the strings in order to produce a martellato effect in *The Phantom Gondolier [Scorpio]*. In addition, the written indications for fingers deviate from traditional fingering suggestions, with guidelines such as "the forefinger and the middle finger of the right hand should be fitted with metal thimbles."[247]

A number of contemporary compositions use traditional piano notation without providing specific performance instructions. Nevertheless, the music is idealistic and virtually unplayable as written; therefore the performer needs to make significant technical and interpretational choices. One characteristic example in this category is Iannis Xenakis' *Evryali*. Xenakis uses chords that exceed a two-octave stretch for each hand. The idealistic character of the composition leaves the choice of notes that should be played to the performer, who can decide fingering suitability based on individual technical abilities and timbral preferences.

Peter Hill explains this procedure:

> [Xenakis] has built into his notation the element of genuine impossibility. In this way he has ensured that each performance will become an attempt at an ideal but unrealizable perfection. The musician is therefore like an athlete, who, in terms of measured achievement, can only aim for improvement, not at some objective goal.[248]

According to Gardner Read in *Compendium of Modern Instrumental Techniques,* extended techniques require a wide range of finger actions from the performer. Read provides an outline of commonly used fingering directions in new music compositions:

1. Pluck the key ("plucked accent") instead of striking it
2. Strike the key forcefully, staccato, and immediately depress it silently
3. Strike the key normally, then raise it (by releasing finger pressure) and depress it again silently before the sound is more than half-dampened
4. Vibrate the key with the finger after striking it (without causing the hammer to hit the string again)
5. Trill on two adjacent black keys with the knuckles
6. As a key is pressed down with one hand, strike the palm simultaneously with the free hand
7. Rapidly wriggle the fingers over the keys; silently jiggle the keys

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[247] George Crumb, "The Phantom Gondolier [Scorpio]" from *Makrokosmos,* vol. I (C.F. Peters, 1974).

[248] Peter Hill, "Xenakis and the Performer," *Tempo,* no. 112, March, 1975, p.19.

8. Lightly strike [the] key with flesh (or with nail) without causing [the] hammer to hit the string
9. Rub the fingernails on the keys (without depressing them) with a constant and uniform movement, as fast as possible
10. Depress the key and strike it with a finger-ring
11. Play on the keys with a heavy wool sock on the left hand
12. Drop a heavy wood stick onto the keys[249]

Overview of modern research on fingering

During the twentieth and early twenty-first centuries, piano music has experienced an unprecedented diversity in compositional styles. In spite of the radical changes in performance demands, the search for the best approach to teaching fingering remains one of the primary concerns of every piano teacher. Educators have concluded that fingering is not just a matter of mechanical predisposition to certain patterns. It affects all aspects of performance, from the comfort of execution to the color of the produced tone. According to William S. Newman, fingering can "profoundly affect memorizing, stage poise, technical mastery, speed of learning, and general security at the piano."[250]

As a result, recent books on the principles of piano playing are not preoccupied with providing standardized fingering for specific patterns. Instead, they provide methods to choose appropriate fingering, taking into account the specific requirements of the piece and matters of human anatomy.

William Stein Newman (1912-2000) published *The Pianist's Problems* in 1950. The fingering section of the book contains virtually no musical examples, but analyzes extensively the important factors that determine an appropriate fingering. Newman bases his fingering principles on the musical context, the desired articulation, and the dynamic intention of each note.

In addition, he provides practical advice, such as adequate notation of fingering (since the overuse of numbers can interfere with note reading), the importance of consistency in fingering during practice sessions, and the necessity for supervision of a young student when a fingering choice is to be made. However, he suggests a particular

[249] Gardner Read, 208-209.
[250] William S. Newman, *The Pianist's Problems* (New York: Da Capo Press, Inc., 1950), 98.

way of fingering which he calls "positioning technique" that aims for the least possible movement of the thumb for every given passage.

> A good way to begin fingering is to take one handful of coverage of notes at a time, regardless of what fingers land on which black or white keys. . . . Within the limits of the technique, the starting finger should be the one that permits the largest number of coming notes to be covered while at least one finger serves as a pivot.[251]

Julien Musafia published *The Art of Fingering in Piano Playing* in 1971. The book is a very comprehensive guide to the principles and different parameters that a pianist should consider before making fingering choices. Musafia refers briefly to pre-Baroque fingering practices and considers the biomechanical fundamentals in all fingering suggestions. He promotes effective fingering that permits the greatest economy of motion, but most importantly provides the contracted muscles with enough time to recover:

> The task of good fingering is to afford as much rest to each finger as possible, by providing recovery time between exertions through judicious distribution of the work between the fingers.[252]

Musafia demonstrates his fingering ideals through an abundance of examples from all periods of keyboard literature. Instead of manufacturing patterns to demonstrate standard fingering procedures, he uses actual musical examples to explore the variety and diversity of fingering options. Each fingered example is accompanied by a short discussion on the procedure and the thought process behind each fingering choice. The various parameters that determine fingering include the symmetry of patterns between the hands, consistency in identically transposed formulas, particular tone color effects, repetitions of notes, and patterns that increase the demand for muscular recovery.

Perhaps the most intriguing part of the book is the discussion on scales and scale patterns. Musafia believes that standardized scale fingering is not always the most effective in terms of economy of motion. Since the movement of passing the thumb

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[251] Ibid., 183-184.
[252] Julien Musafia, *The Art of Fingering in Piano Playing* (New York: MCA Music, 1971), 2.

under the fourth finger creates maximum contraction, Musafia suggests fingering that minimizes that circumstance, especially when both fingers are playing on white keys.

> We can find new and better fingerings for these scales through those deductions which state that turning the thumb after a black key is easier than turning after a white key, and that turning the thumb after a finger of low numeral is easier than turning after a finger of high numeral. Thus the scale of C minor harmonic can avoid the thumb turning after 4 on B by placing 4 on E-flat and 3 on A-flat. With this fingering the thumb will play only after black keys.[253]

The role of the thumb is examined in detail throughout the book in various circumstances, such as its function in hand extension and in finger substitution. In addition, many other fingering issues are discussed including legato octave fingering, glissando, fingering distribution in polyphonic pieces, and arpeggios.

Penelope Roskell published *The Art of Piano Fingering: A New Approach to Scales and Arpeggios* in 1996. The book rationalizes the fundamentals of scale fingering from both a traditional and an alternative point of view. The book is divided into chapters that discuss separate patterns: minor and major scales, double notes, arpeggios, octaves, and so forth.

At the beginning of each chapter some general fingering principles are mentioned, followed by examples that illustrate various points. These principles are very similar to the basic eighteenth-century fingering ideas. The discussion of each fingering issue aims to justify particular fingering decisions. The suggested fingering is for the most part traditional since only at the end of each chapter is some alternative fingering presented.

Throughout the twentieth century scholars have demonstrated an interest in studying fingering in accordance with human anatomy and the basic principles of contemporary music education. Harry Spangler's thesis, "An Historical and Experimental Study of Some of the Motor Aspects of Pianoforte Technique" from 1933, associates the technical issues of the pianist with the fields of neurology and motor performance. The thesis includes a brief historical survey of pianoforte technique, beginning with C.P.E. Bach and ending with Tobias Matthay.

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[253] Ibid., 6.

The main part of Spangler's thesis concentrates on the neural impulses upon which muscular control is based.[254] The structure of muscles and the stages of voluntary and involuntary muscle motions and contractions are analyzed thoroughly. Although this study does not contribute directly to the subject of fingering, it explains the physiological and neurological factors that make any finger motion possible.

Vera Bernice Wright completed a thesis on "The Degree to which Fingering Aids in Developing a Piano Technique and in Acquiring a Knowledge of the Keyboard" in 1938. This is an experimental study aiming to compare two groups of children who were asked to repeat the same pattern and transpose it to various keys. One group was taught a particular fingering sequence, while the other was simply taught a musical pattern.

The results of the study prove that the children who were taught the fingering sequence could play the pattern more accurately. They were also able to transpose the pattern very easily, while the other group encountered major problems in the transposition.[255] The group with fixed fingering required more time to learn a pattern but the effectiveness in the use of consistent pre-calculated fingering was overwhelming.

Barbara Ann Cornehl's "A Resume and Bibliography of Piano Fingering Material" of 1956 summarizes the fingering ideas of twentieth-century piano teachers.[256] This research is based principally on articles and interviews from periodicals. Finally, Robert Joseph Roux's 1980 D.M.A. treatise, "A Methodology of Piano Fingering," examines in detail "the structure of the hand and its relationship to the topography of the keyboard."[257] Based on hand structure, Roux analyzes selected musical examples and provides one or more solutions. His fingering suggestions are not formed on a strictly anatomical basis, but other parameters such as musical content are analyzed and taken into consideration.

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[254] Harry Spangler, "An Historical and Experimental Study of Some of the Motor Aspects of Pianoforte Technique" (Master's thesis, University of North Dakota, 1933), 24.

[255] Vera Bernice Wright, "The Degree to which Fingering Aids in Developing a Piano Technique and in Acquiring a Knowledge of the Keyboard" (Master's thesis, Indiana University, 1938), 27.

[256] Barbara Ann Cornehl, "A Resume and Bibliography of Piano Fingering Material" (Master's thesis, Montana State University, 1958).

[257] Robert Joseph Roux, "A Methodology of Piano Fingering" (D.M.A. treatise, University of Texas at Austin, 1980), 12.

Discussion

From the beginning of the twentieth century until the present time, the issue of fingering instruction is seldom treated as an independent technical aspect. The exception to this would be those methods that explore possibilities of new and unconventional ways of teaching standard pianistic patterns such as scales and arpeggios. Discussions on fingering by the major pedagogues and researchers of the twentieth century are consistently associated with the investigation of the muscular and neurological coordination of the playing apparatus.

Specific practical suggestions are provided only in cases where composers require certain effects, or researchers explore particular technical and interpretational difficulties of various pieces from the keyboard literature. Piano teachers seem to be more concerned with providing scientific and philosophical foundations that should determine an effective fingering than with suggesting fixed fingering solutions.

The change of approach to fingering can be attributed partially to a growing awareness of the complexity of the muscular mechanism and partially to the contemporary educational philosophy of teaching basic principles instead of dogmas. This approach requires more preparation and experimentation from the individual pianist than do the prescribed formulas.

Nevertheless, apart from the purely technical demands of any given passage, the chosen fingering needs to meet a wide range of individual requirements: personal ease during performance of the passage, minimum muscular tension, and use of individual finger characteristics in order to achieve desired tone colors. As Newman strongly states, "the choice of and adherence to a fingering on a keyboard instrument can make or break a piece."[258]

[258] Newman, 98.

CONCLUSION

A historical overview of keyboard fingering instructional material from the last five centuries reveals a remarkable diversity due to the varying degrees of systematization of the material itself, as well as to fundamentally different pedagogical methodologies. These differences are related to the keyboard repertoire, performance practices and historical and sociological contexts of different eras.

The significance of performance practice traditions regarding fingering was predominantly evident from the first available keyboard tutors until the end of the Baroque era. This was the only time in keyboard history that fingering practices and teachings varied considerably among different European countries, thus reflecting individuality in compositional and interpretational approaches.

Despite the differences in fingering choices, all of the sources from this early period represent humanistic and rationalistic ideals. Even though scientific anatomical knowledge of the hand was limited, keyboard instructors based their teaching principles on the different potential of fingers due to their varying length and strength. They also acknowledged the need for a systematic organization of their teaching practices in order to achieve effectiveness. As a result, the tutors began to address a number of issues related to keyboard playing, thus forming the principal concepts of keyboard instruction books.

The paired pre-Baroque and Baroque keyboard fingering was designed to suit the character of the keyboard repertoire of the time, which was performed predominantly on harpsichord. Therefore, it has limited application to the technique of a modern pianist. On the other hand, in recent decades the revival of period instruments and authentic performance practices of early music has become increasingly popular. In this respect, careful study of early fingering and its effect on articulation is an essential tool for the better understanding and performance of early music.

The period of Enlightenment initiated a fundamental change in both the level of organization of keyboard teaching material and the specific fingering instructions. This

period reflected a transition from the Baroque to the Classical style and technique. Treatises of the time included provisions for keyboardists trained in the paired fingering system; nevertheless, the increasing predominance of the fortepiano and the technical requirements of the keyboard repertoire of the Classical period led to a technique that took advantage of the full potential of the human hand.

As the fortepiano continued to advance as an instrument, gaining range and dynamic diversity, the technical demands on keyboard players increased. The scientific approach in the systematization of keyboard instruction methodologies that had begun in the mid-eighteenth century reached its peak in the first three decades of the nineteenth. Attributing a pivoting role to the thumb changed the philosophy of fingering pedagogy. In addition, a considerable change in the targeted audience for keyboard instruction material took place.

Writers from the mid-eighteenth to mid-nineteenth centuries addressed their advice not only to a selected group of students of predominantly aristocratic descent, as was the case in previous centuries, but also to aspiring students who came from the middle class and desired musical education but who were not fortunate enough to study with highly trained teachers. As a result, the methods were more systematic and detailed than ever before.

Furthermore, writers from the first half of the nineteenth century aspired to provide piano students with a technical system that would ensure the effective training of fingers in order to achieve the utmost finger dexterity, a requirement for ideal virtuosity. In order to achieve this goal, keyboard instruction books gradually became method books that contained an abundance of exercises based primarily on scale and scale-like patterns.

This “finger gymnastic” approach, despite its technical straightforwardness, was by itself not sufficient to respond to the demands of Romantic music, which was highly virtuosic, yet predominantly emotional. Famous teachers of the time, afraid of producing pianists with sufficient technical skills who lacked quality in the tone production, were very reluctant to write instruction books.

As a result, the second half of the nineteenth century saw a remarkable decrease in the number and quality of piano tutors. The few existing books continued the tradition of the finger gymnastic approach without any new insight into fingering. The most

accomplished teachers of the Romantic era left only sketches of instructional material, and the majority of information regarding their teachings, in particular their views on fingering, came primarily from their students' accounts.

From these accounts it is evident that the fundamental factors affecting fingering decisions were the desired tone and the creation of special effects. After a century of intense exercises that aimed to train fingers to be equal and independent, teachers from the Romantic period returned to the Baroque idea that all fingers by nature do not have the same potential. Therefore, instead of exhaustively training them to achieve independence, they suggested innovative fingering that took advantage of the different tone color that each finger produced naturally.

The major pianists and teachers from the beginning of the twentieth century came from the Romantic tradition; therefore a significant amount of keyboard instruction books reflected the Romantic ideals. Parallel to this tradition though, the first keyboard instruction books that demonstrated a thorough knowledge of the human muscular system made their appearance early in the century.

This new generation of keyboard instruction books incorporated detailed scientific anatomical knowledge of the muscular and neurological coordination of the playing apparatus. The heavier action of the piano and the increasingly technically demanding piano repertoire made this scientific knowledge an imperative tool for teachers and pianists in order to understand the playing mechanism and make decisions on issues such as fingering.

The anatomical and physiological orientation of the majority of twentieth century keyboard instruction books restricted them from addressing performance and technical issues in an empirical manner. Instead of prescribing fingering formulas and rules, these manuals aimed to establish an awareness of the complexity of the playing apparatus. The performer was urged to make fingering decisions based on individual muscular abilities, tension and relaxation issues, and desired tone colors. Finally, performers of contemporary piano music occasionally have to use unconventional fingerings and extended piano techniques in order to meet the requirements of new music.

Fingering has always been one of the major topics of keyboard instruction manuals. Even though teachers and pedagogues have considered the adequate choice of

fingering an indispensable technical tool for piano playing, the inherent difficulties of fingering decisions have created a wide variety of fingering principles and methods. A historical overview of fingering ideas reveals the whole range of teaching approaches: from dry citations of rules to hidden suggestions in imaginary dialogues, and from prescribed fingered technical patterns to compositions with strategically placed fingerings aimed to change a particular tone color.

In spite of the formal differences of written fingering material, the generating force behind fingering principles has not changed throughout time. The authors of the examined tutors aimed to teach fingering that helped students play more effectively according to the individual musical demands, with the least amount of muscular effort and tension. The philosophical changes in teaching fingering simply reflect the social, historical, musical and pedagogical changes that are vital forces of every creative aspect of human life.

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BIOGRAPHICAL SKETCH

Athina Fytika was born and raised in Athens, Greece. She started taking piano lessons at the age of eight and received her Diploma of Piano Performance from Contemporary Conservatory of Thessaloniki. She studied piano with Chrissi Partheniadi and Domna Evnouhidou, advanced theory with Giannis Avgerinos and counterpoint with Kostantinos Siebis. She also received a Bachelors degree in Geological Sciences from Aristotle University of Thessaloniki, Greece. Upon completion of her undergraduate studies, she moved to the United States. She received her Masters degree in Piano Performance from Florida State University, where she studied piano with Mr. Leonard Mastrogiacomo. During her studies as a doctoral student in Piano Performance at Florida State University, she studied piano with Dr. Carolyn Bridger and harpsichord with Dr. Karyl Louwenaar. Her teaching experience includes music instruction in elementary schools, as well as piano instruction in both conservatory and college settings. She is currently an adjunct piano instructor at Gulf Coast Community College in Panama City, Florida.